AM62x STARTER KIT EVM

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REV	E3
VER	0.07

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TEXAS INSTRUMENTS

MISTRAL

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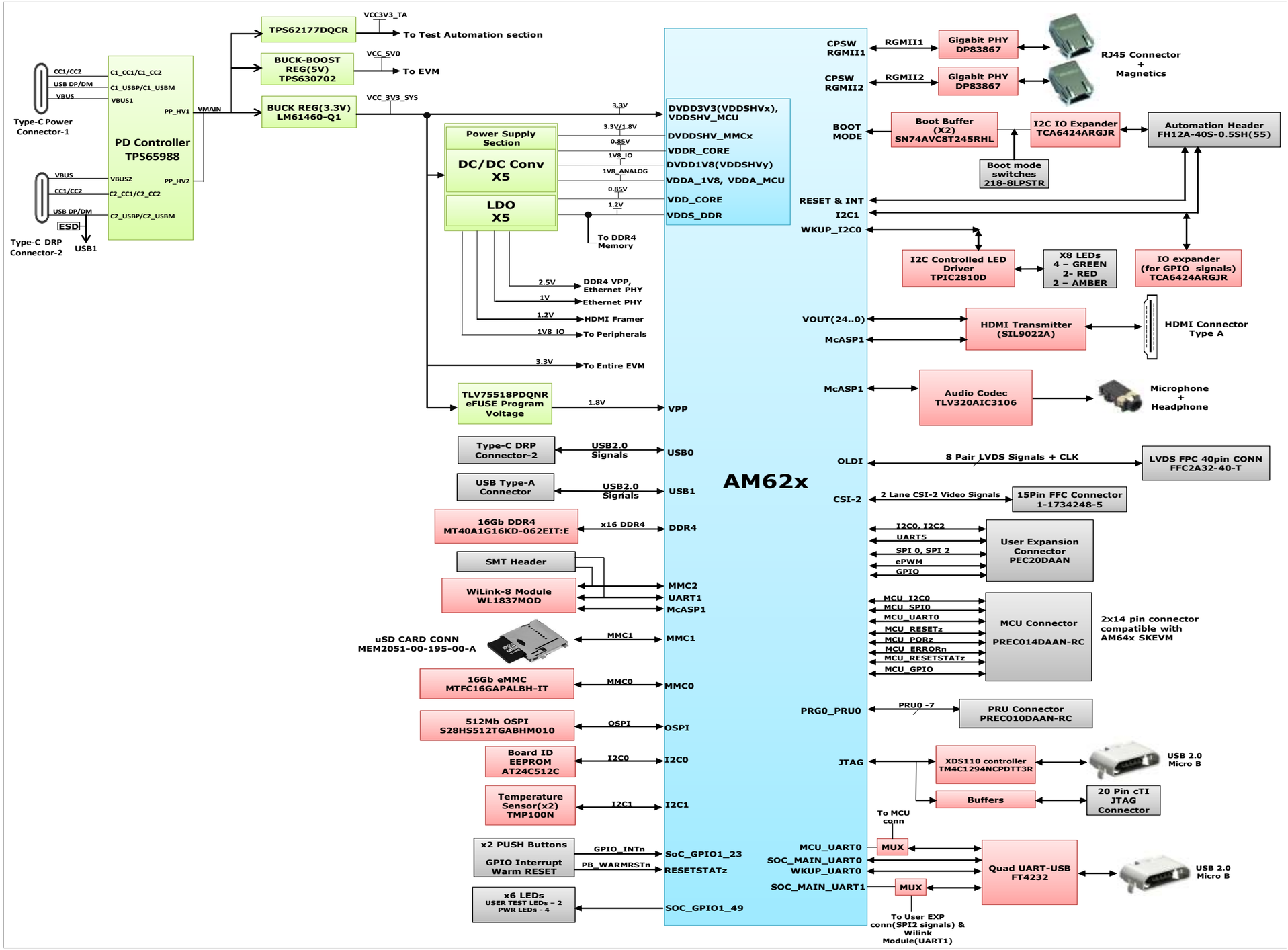
REVISION HISTORY

VER #	DATE	DESCRIPTION OF CHANGES	AUTHOR	REVIEWED BY	APPROVED BY
0.01	16 MARCH 2022	Drafted from E2 Schematics Changed the LVDS Connector J21 Pinout	Mistral Design Team		
0.02	25 MARCH 2022	Updated the UART1_BT signals connection for Buffer U3 and Buffer U60 connections are modified	Mistral Design Team		
0.03	31 MARCH 2022	Added isolation buffers for XDS_TA signals from TM4C129 microcontroller Removed the 0E res option for VCC3V3_TA supply	Mistral Design Team		
0.04	11 APRIL 2022	Connected the VCC of U18 ESD device on MMC1_SDCD line to VCC_3V3_SYS rail	Mistral Design Team		
0.05	13 APRIL 2022	Connected the I2C2_IRQ from PD controller to IO Expander U70.18 Terminated the TEST_GPIO2 signal to a TP Tripad RES stuffing option(R697 and R698) provided for P14 GPIO from IO EXP to either HDMI and OLDI reset	Mistral Design Team		
0.06	19 APRIL 2022	U60 I/O connections swapped to ease routing U76 PS/SYNC pin connected to VMAIN through 10K Res	Mistral Design Team		
0.07	22 APRIL 2022	Changed U23 and U25 I2C slave addressed to 0x4D and 0x47	Mistral Design Team		

BLOCK DIAGRAM AM62x SKEVM



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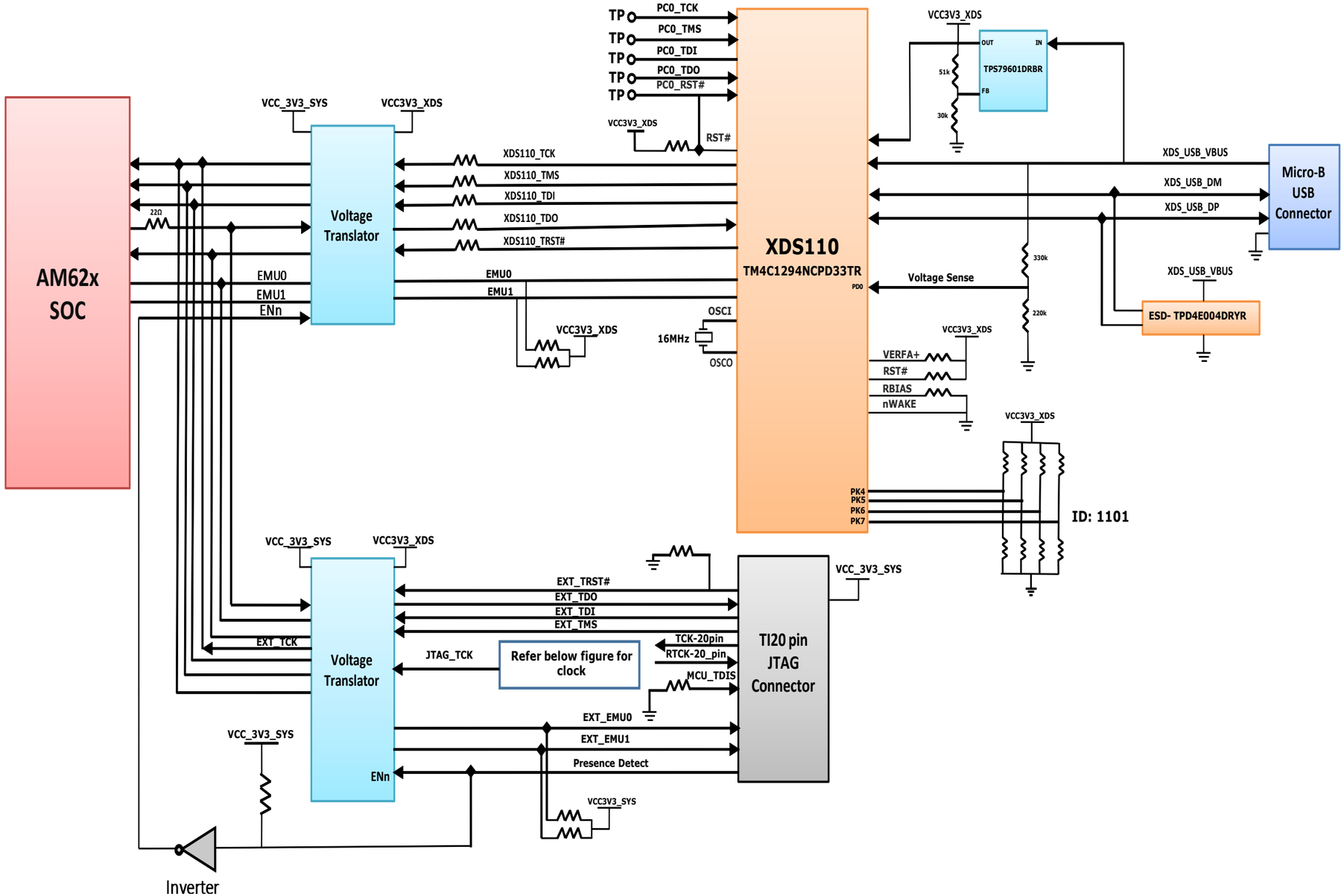
BLOCK DIAGRAM_XDS110



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Title	BLOCK DIAGRAM_XDS110	
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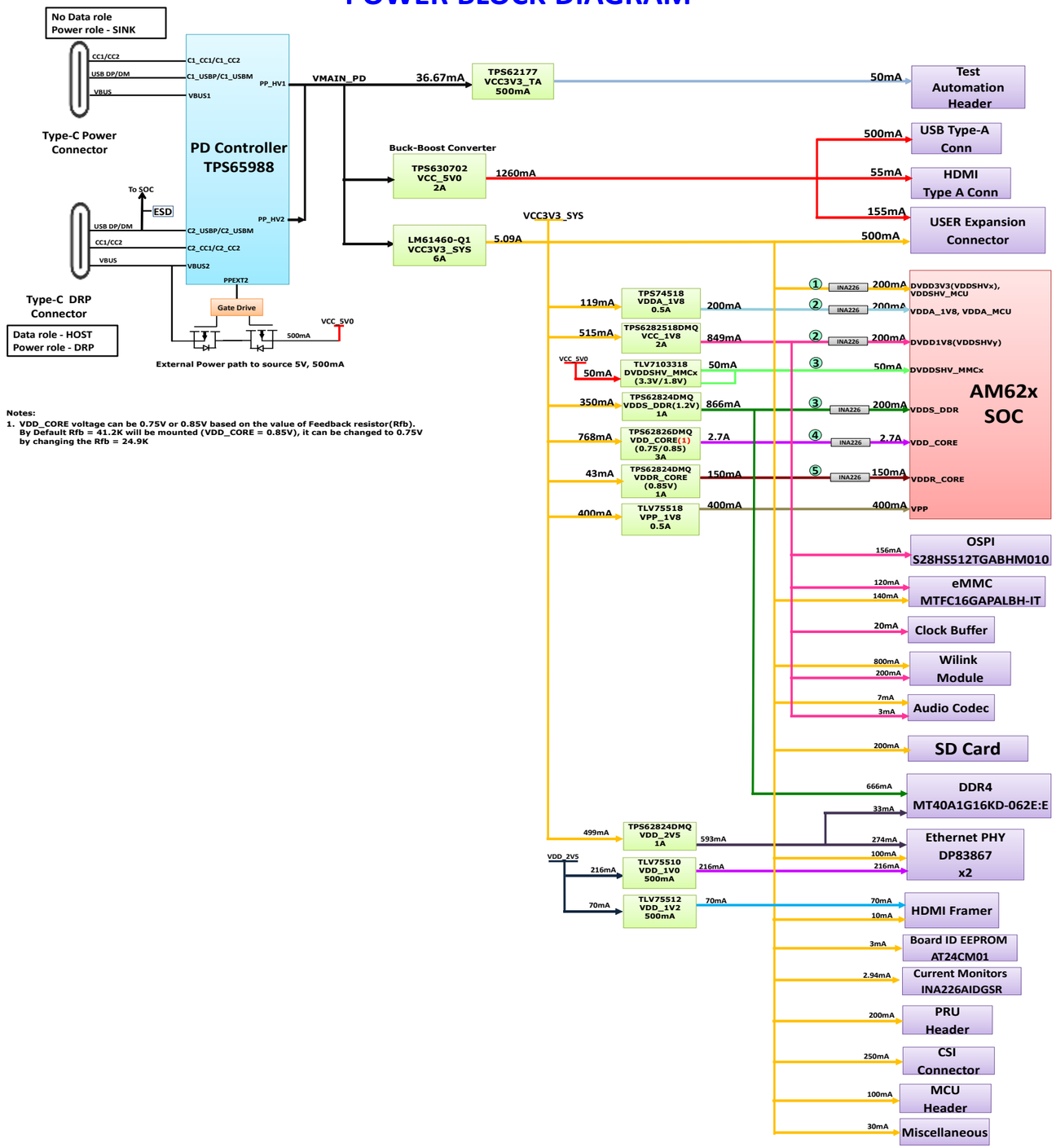
POWER BLOCK DIAGRAM



Notes:

1. VDD_CORE voltage can be 0.75V or 0.85V based on the value of Feedback resistor(Rfb). By Default Rfb = 41.2K will be mounted (VDD_CORE = 0.85V), it can be changed to 0.75V by changing the Rfb = 24.9K

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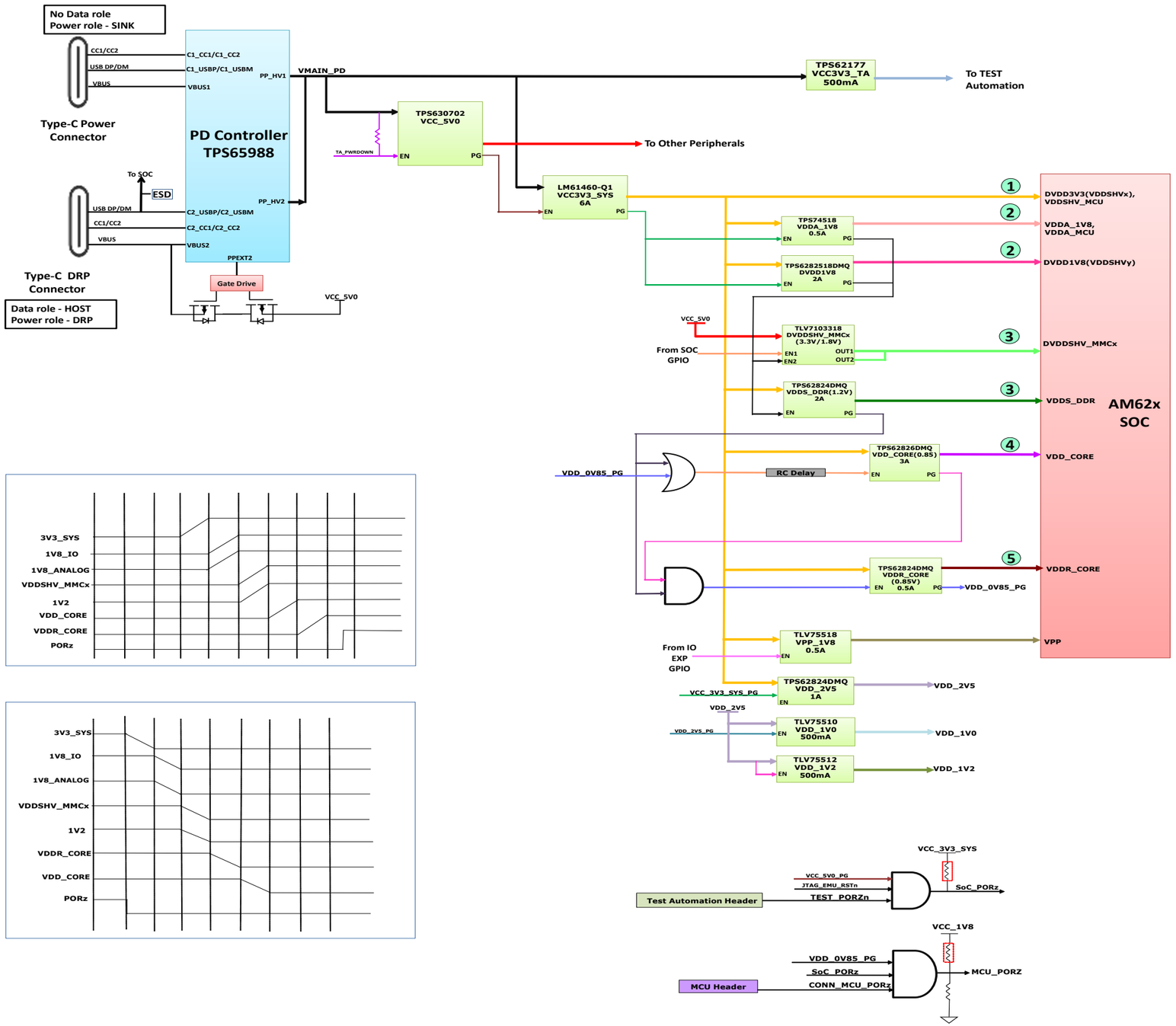
POWER SEQUENCE



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Title	POWER SEQUENCE		
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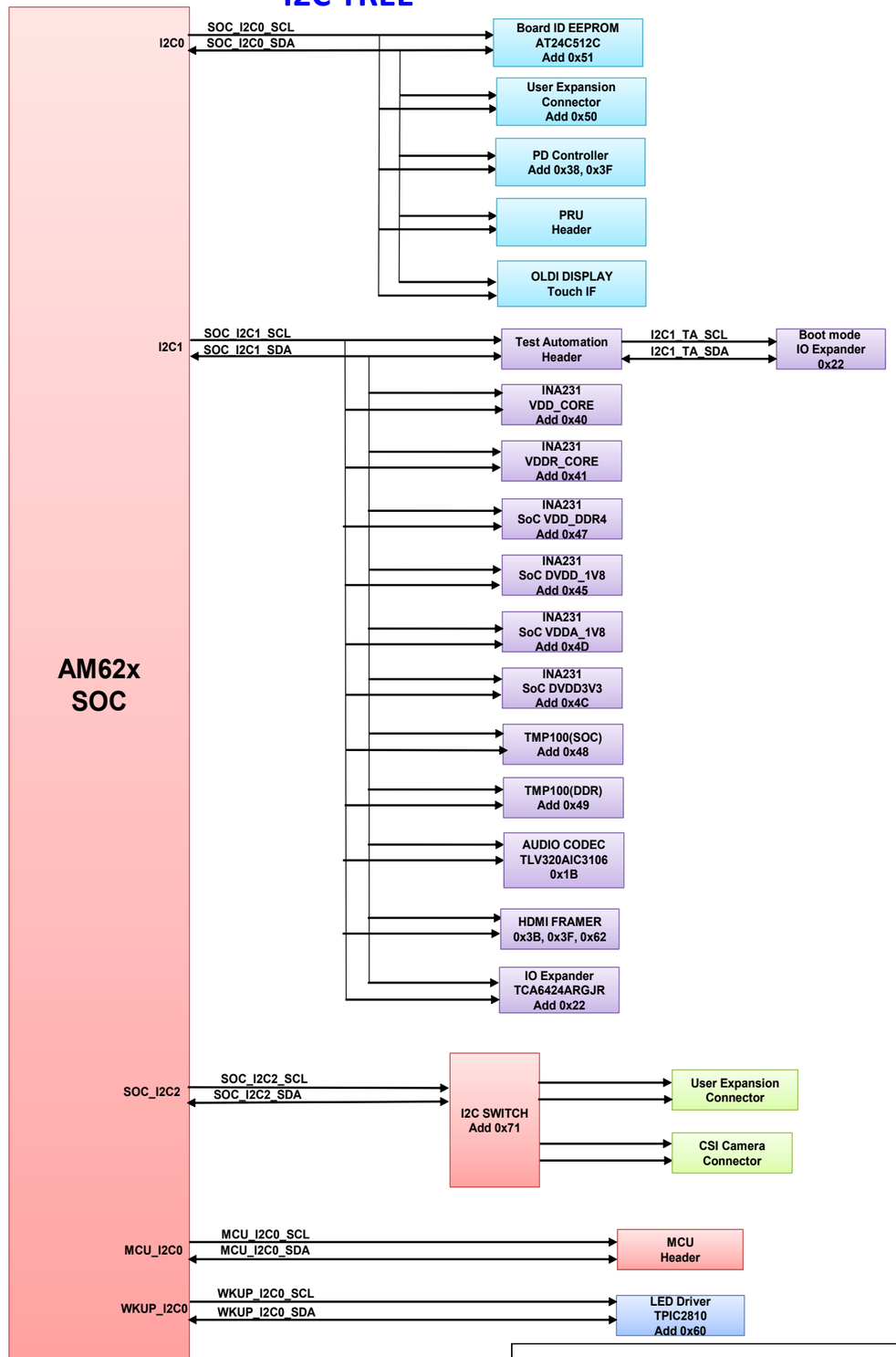
I2C TREE



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Title	I2C TREE	
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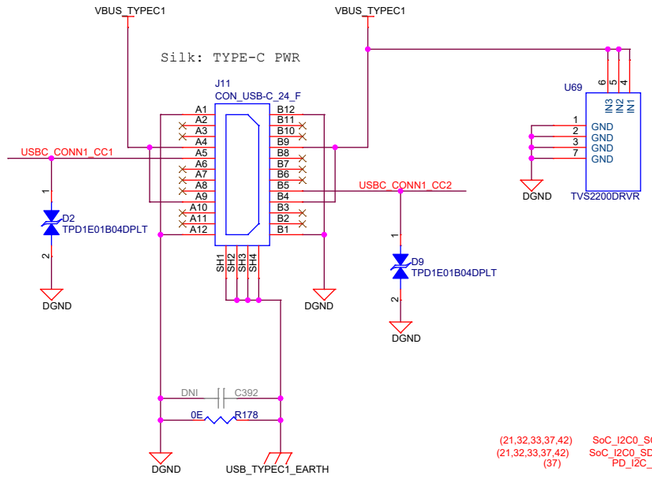
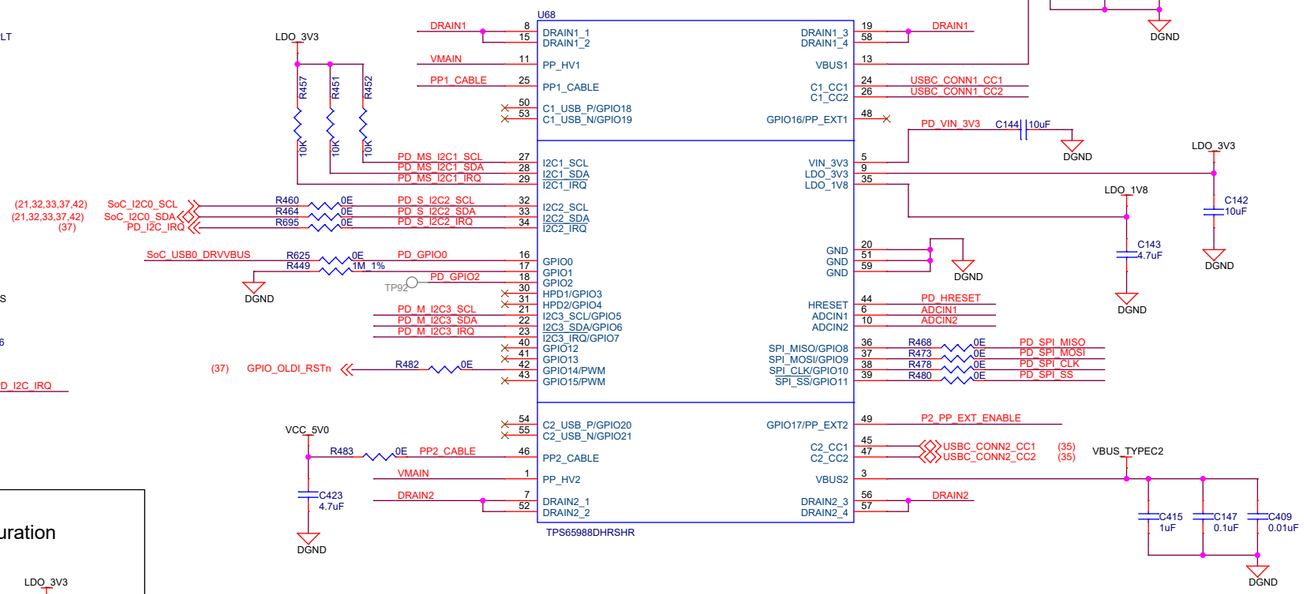
GPIO MAPPING TABLE

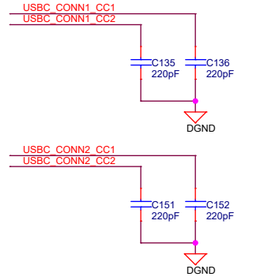
SL NO.	GPIO DESCRIPTION	GPIO NETNAME	FUNCTIONALITY	GPIO USED	SOC MUXED SIGNAL NAME	DIRECTION WITH RESPECT TO CONTROL	DEFAULT STATE	ACTIVE STATE	VOLTAGE DOMAIN ON SOC SIDE	VOLTAGE CONNECTED ON SKEVM
1	Enable for WLAN Interface	WLAN_EN	ENABLE	GPIO0_71	MMC2_SDCD	OUTPUT	LOW	HIGH	VDDSHV6	SoC_DVDD1V8
2	WLAN Interrupt	WLAN_IRQ	INTERRUPT	GPIO0_72	MMC2_SDWP	INPUT	HIGH	LOW	VDDSHV6	SoC_DVDD1V8
3	Enable for BT Interface	BT_EN_SOC	ENABLE	MCU_GPIO0_1	MCU_SPI0_CS0	OUTPUT	LOW	HIGH	VDDSHV_MCU	SoC_DVDD3V3
	CPSW Ethernet PHY Interrupt	CPSW_RGMII_INTn/PRU_INTn	INTERRUPT	GPIO1_31	EXTINTn	INPUT	HIGH	LOW	VDDSHV0	SoC_DVDD3V3
4	PRU Connector Interrupt									
5	OSPI Reset Control GPIO	GPIO_OSPI_RSTn	RESET	GPIO0_12	OSPI0_CSn1	OUTPUT	HIGH	LOW	VDDSHV1	SoC_DVDD1V8
6	OSPI Interrupt	OSPI_INTn	INTERRUPT	GPIO0_13	OSPI0_CSn2	INPUT	HIGH	LOW	VDDSHV1	SoC_DVDD1V8
7	SD Card IO Voltage Select	VSEL_SD	ENABLE	GPIO0_31	GPMC0_CLK	OUTPUT	LOW	HIGH	VDDSHV3	SoC_DVDD3V3
8	IO Expander Interrupt	GPIO1_23_INTn	INTERRUPT	SoC_GPIO1_23	UART0_RTSN	INPUT	HIGH	LOW	VDDSHV0	SoC_DVDD3V3
9	User Interrupt Push Button/ TEST GPIO1 from Test Automation Connector									
10	User Test LED 1	SOC_GPIO1_49	GPIO	GPIO1_49	MMC1_SDWP	OUTPUT	LOW	HIGH	VDDSHV0	SoC_DVDD3V3
	IO EXPANDER - 01									
1	CPSW Ethernet PHY-2 Reset Control GPIO	GPIO_CPSW2_RST	RESET	IO EXPANDER - P00		OUTPUT	HIGH	LOW	VDDSHV0	SoC_DVDD3V3
2	CPSW Ethernet PHY-1 Reset Control GPIO	GPIO_CPSW1_RST	RESET	IO EXPANDER - P01		OUTPUT	HIGH	LOW	VDDSHV0	SoC_DVDD3V3
3	PRU Board Detection	PRU_DETECT	DETECTION	IO EXPANDER - P02		INPUT	HIGH	LOW	VDDSHV0	SoC_DVDD3V3
4	SD Card Load Switch Enable	MMC1_SD_EN	ENABLE	IO EXPANDER -P03		OUTPUT	HIGH	LOW	VDDSHV0	SoC_DVDD3V3
5	SOC eFuse Voltage(VPP=1.8V) Regulator Enable	VPP_LDO_EN	ENABLE	IO EXPANDER - P04		OUTPUT	LOW	HIGH	VDDSHV0	SoC_DVDD3V3
6	EXP CONN 3.3V Power Switch Enable	EXP_PS_3V3_EN	ENABLE	IO EXPANDER - P05		OUTPUT	LOW	HIGH	VDDSHV0	SoC_DVDD3V3
7	EXP CONN 5V Power Switch Enable	EXP_PS_5V0_EN	ENABLE	IO EXPANDER - P06		OUTPUT	LOW	HIGH	VDDSHV0	SoC_DVDD3V3
8	EXP CONN HAT Board Detection	EXP_HAT_DETECT	DETECTION	IO EXPANDER - P07		INPUT	HIGH	LOW	VDDSHV0	SoC_DVDD3V3
9	Audio Codec Reset Control GPIO	GPIO_AUD_RSTn	DETECTION	IO EXPANDER - P10		INPUT	HIGH	LOW	VDDSHV0	SoC_DVDD3V3
10	eMMC Reset control GPIO	GPIO_eMMC_RSTn	RESET	IO EXPANDER - P11		OUTPUT	HIGH	LOW	VDDSHV0	SoC_DVDD3V3
11	UART1 FET Switch and Buffer Enable signal	UART1_FET_BUF_EN	ENABLE	IO EXPANDER - P12		OUTPUT	HIGH	LOW	VDDSHV0	SoC_DVDD3V3
12	Enable for Wilink Level Translators	WL_LT_EN	ENABLE	IO EXPANDER - P13		OUTPUT	LOW	HIGH	VDDSHV0	SoC_DVDD3V3
13	HDMI Transmitter Reset Control GPIO	GPIO_HDMI_RSTn	RESET	IO EXPANDER - P14		OUTPUT	HIGH	LOW	VDDSHV0	SoC_DVDD3V3
14	Raspberry Pi Camera CSI0 GPIO1	CSI_GPIO1	INPUT/OUTPUT	IO EXPANDER - P15		NA	NA	NA	VDDSHV0	SoC_DVDD3V3
15	Raspberry Pi Camera CSI0 GPIO2	CSI_GPIO2	INPUT/OUTPUT	IO EXPANDER - P16		NA	NA	NA	VDDSHV0	SoC_DVDD3V3
16	PRU Power Switch Enable	PRU_3V3_EN	ENABLE	IO EXPANDER - P17		OUTPUT	LOW	HIGH	VDDSHV0	SoC_DVDD3V3
17	HDMI Interrupt	HDMI_INTn	INTERRUPT	IO EXPANDER - P20		INPUT	HIGH	LOW	VDDSHV0	SoC_DVDD3V3
18	PD Controller Interrupt	PD_I2C_IRQ	INTERRUPT	IO EXPANDER - P21		INPUT	HIGH	LOW	VDDSHV0	SoC_DVDD3V3
19	MCASP1 FET Switch Enable	MCASP1_FET_EN	ENABLE	IO EXPANDER - P22		OUTPUT	LOW	HIGH	VDDSHV0	SoC_DVDD3V3
20	MCASP1 Level Translator buffer for BT Enable	MCASP1_BUF_BT_EN	ENABLE	IO EXPANDER - P23		OUTPUT	LOW	HIGH	VDDSHV0	SoC_DVDD3V3
21	MCASP1 FET Switch select pin status	MCASP1_FET_SEL	GPIO	IO EXPANDER - P24		INPUT	LOW	HIGH	VDDSHV0	SoC_DVDD3V3
22	SOC UART1 FET Switch Select	UART1_FET_SEL	SELECT	IO EXPANDER - P25		OUTPUT	LOW	HIGH	VDDSHV0	SoC_DVDD3V3
23	OLDI Display Touch Interrupt	TS_INT#	INTERRUPT	IO EXPANDER - P26		INPUT	HIGH	LOW	VDDSHV0	SoC_DVDD3V3
24	User Test LED 2	IO_EXP_TEST_LED	GPIO	IO EXPANDER - P27		OUTPUT	LOW	HIGH	VDDSHV0	SoC_DVDD3V3

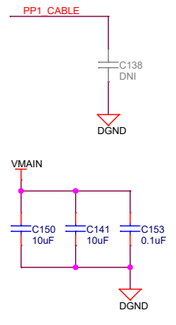
USB TYPE-C POWER

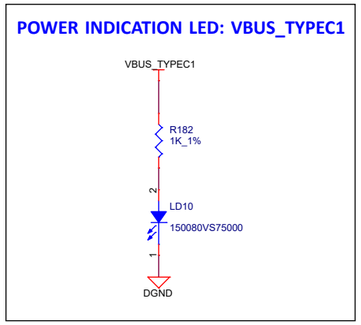


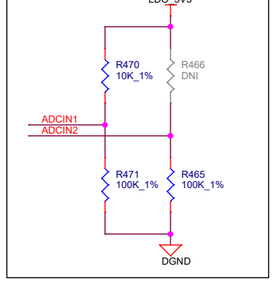
TYPE-C DUAL PD CONTROLLER





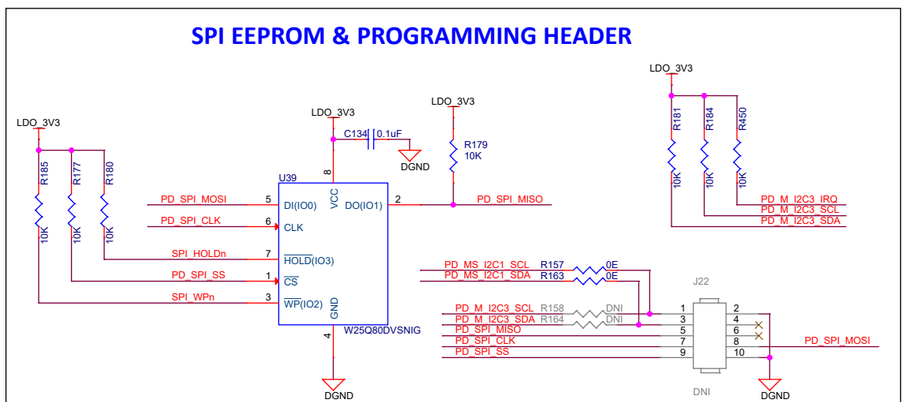




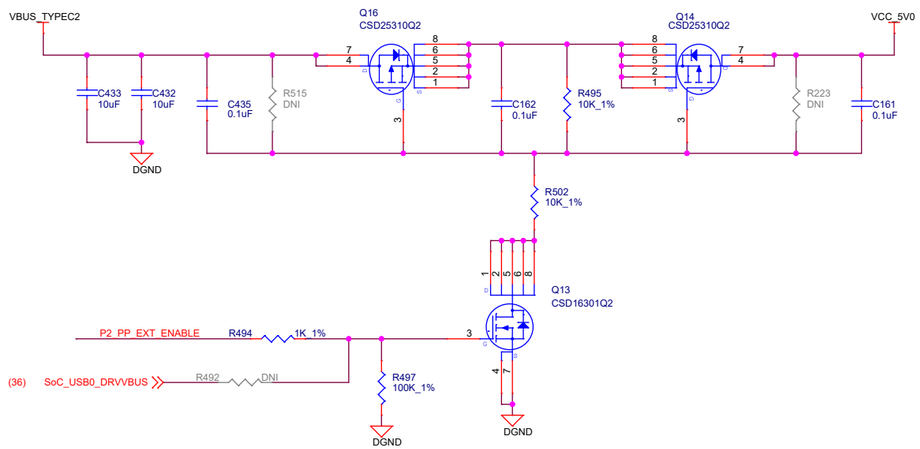


I2C Slave Address	Port1	Port2
I2C2(Default)	0x38	0x3F
I2C1	0x20	0x24

SPI EEPROM & PROGRAMMING HEADER



EXTERNAL POWER PATH FOR SOURCING, 5V/0.5A



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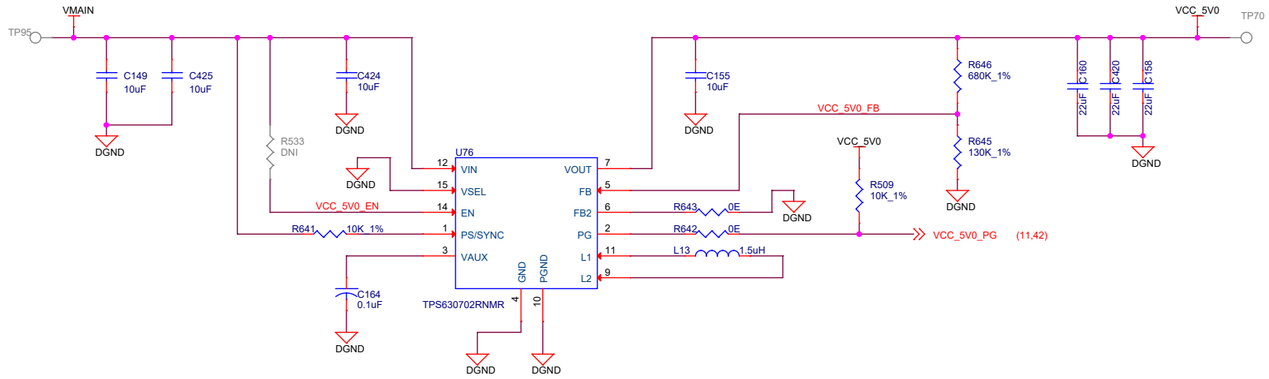
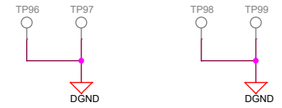
PERIPHERAL POWER SUPPLY-1



Power Cycle control from Test Automation

GROUND TEST POINTS



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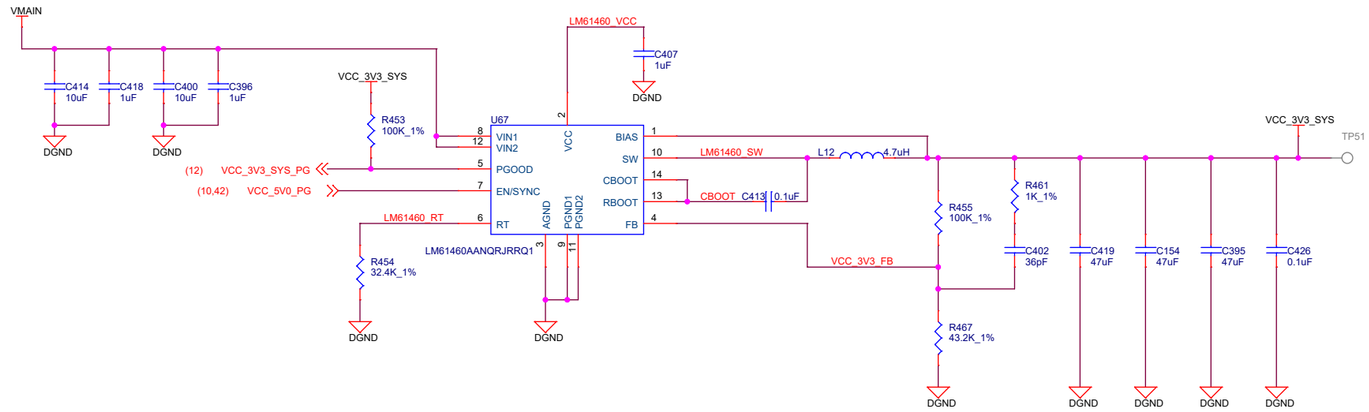
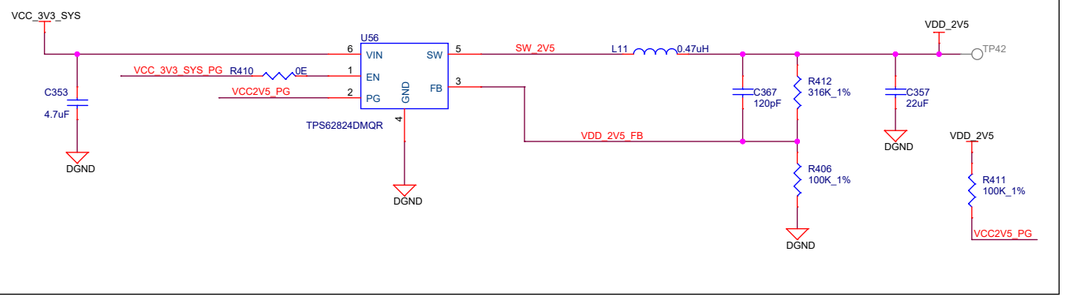
PERIPHERAL POWER SUPPLY-2

3.3V, 6.0AMPS SUPPLY

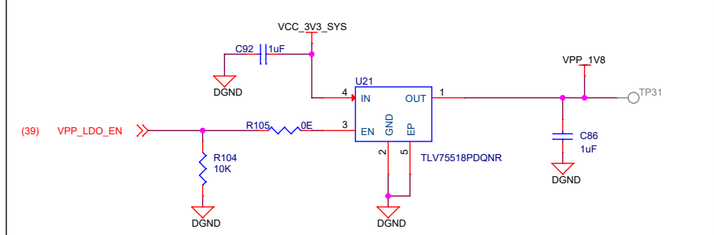
VinMin = 4.5V
VinMax = 24V
Vout = 3.3V @ 6A



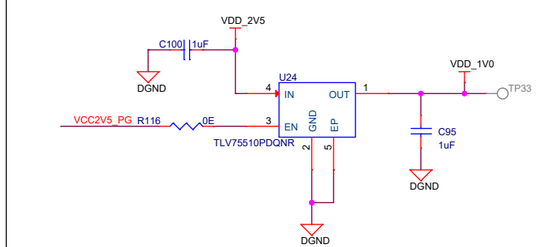
2.5V, 1.0AMPS SUPPLY



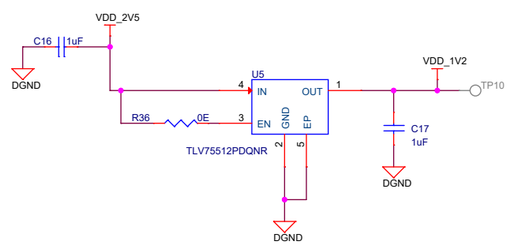
1.8V VPP, 0.5AMPS SUPPLY



1.0V, 0.5AMPS SUPPLY



1.2V, 0.5AMPS SUPPLY



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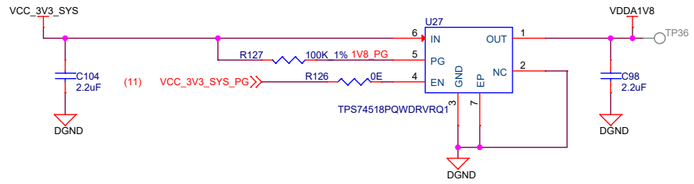
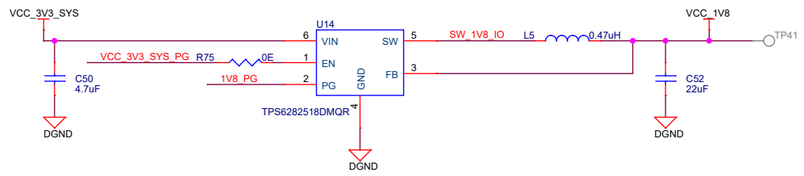
SOC POWER SUPPLY

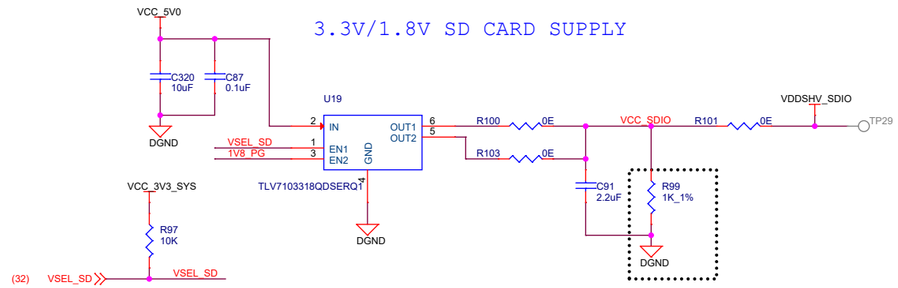
1.8V ANALOG, 0.5 AMPS SUPPLY



1.8V IO, 2.0 AMPS SUPPLY



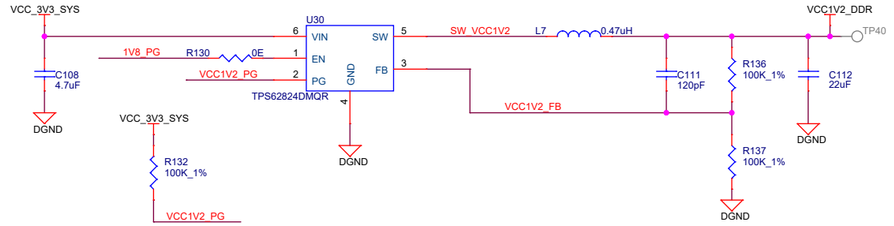
3.3V/1.8V SD CARD SUPPLY



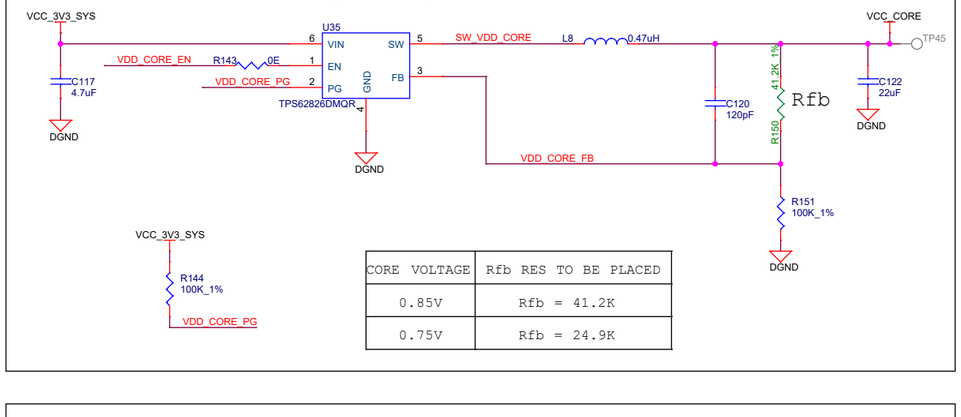
Note: Resistor is to 'bleed' off voltage.

VSEL_SD	SD VOLTAGE
HIGH(DEFAULT)	3.3V
LOW	1.8V

1.2V , 1.0 AMPS SUPPLY

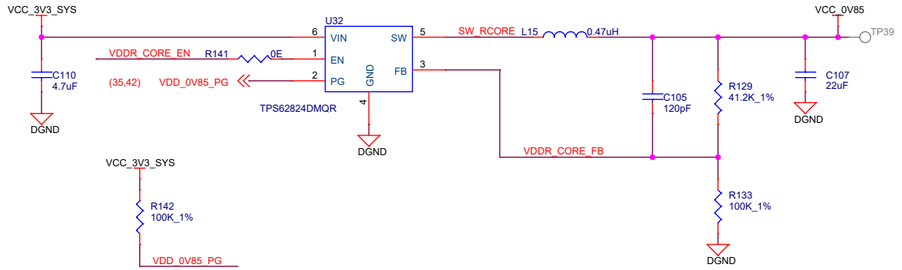


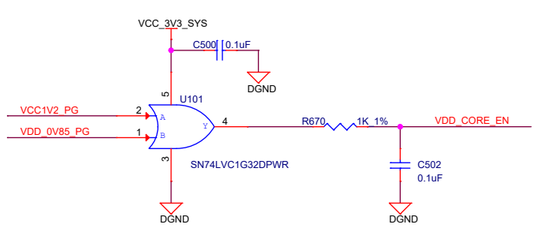
0.75V/0.85V , 3.0 AMPS SUPPLY

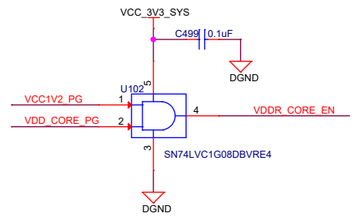


CORE VOLTAGE	Rfb RES TO BE PLACED
0.85V	Rfb = 41.2K
0.75V	Rfb = 24.9K

0.85V, 0.5 AMPS SUPPLY







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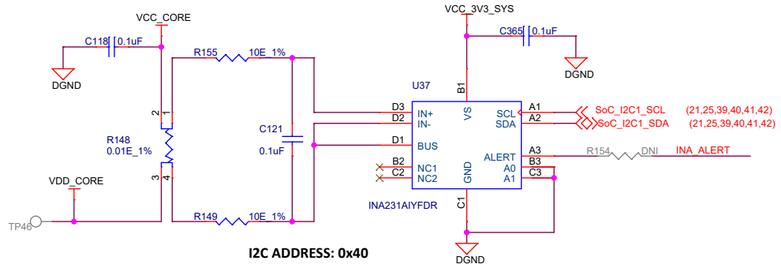
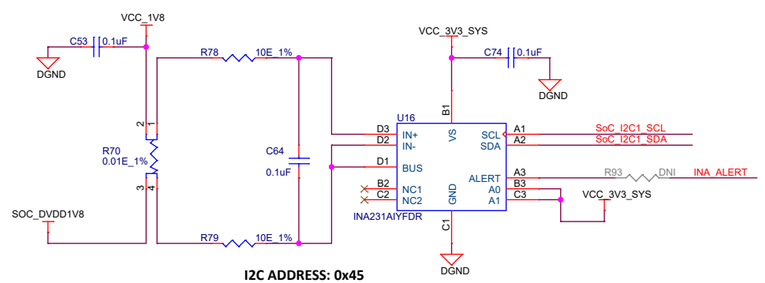
CURRENT MONITORING DEVICES

VDD_CORE



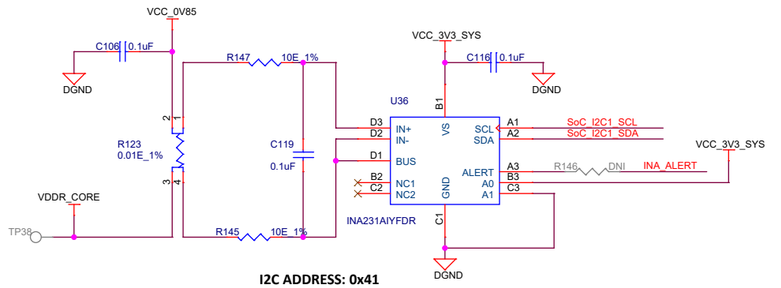
I2C ADDRESS: 0x40

SoC_DVDD1V8



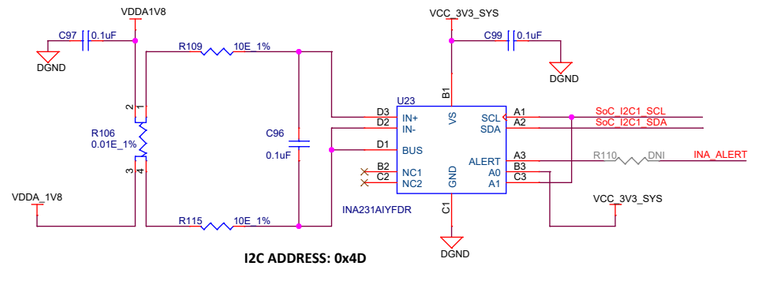
I2C ADDRESS: 0x45

VDDR_CORE



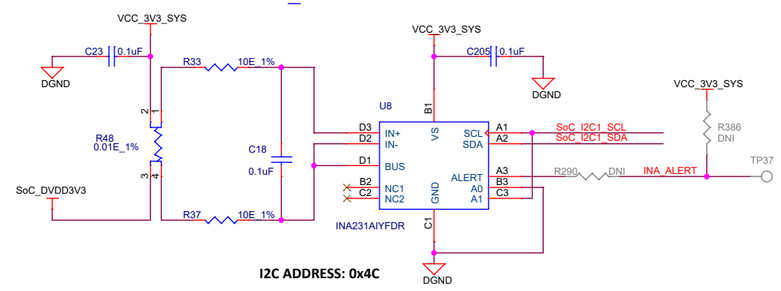
I2C ADDRESS: 0x41

VDDA_1V8



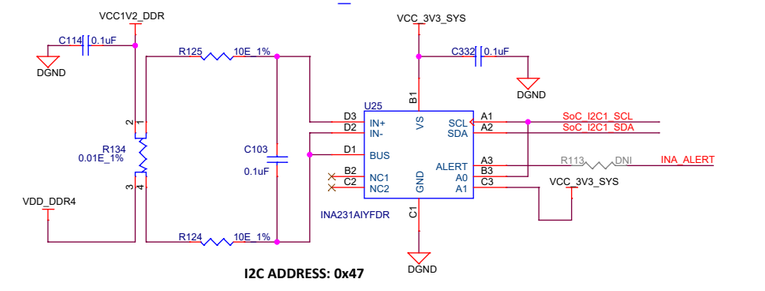
I2C ADDRESS: 0x4D

SoC_DVDD3V3



I2C ADDRESS: 0x4C

VDD_DDR4



I2C ADDRESS: 0x47

INA I2C SLAVE ADDRESS		
POWER SOURCE	SUPPLY NET	SLAVE ADDRESS (IN HEX)
VCC_CORE	VDD_CORE	40
VCC_0V85	VDDR_CORE	41
VCC_3V3_SYS	SoC_DVDD3V3	4C
VCC_1V8	SoC_DVDD1V8	45
VDDA1V8	VDDA_1V8	4D
VCC1V2_DDR	VDD_DDR4	47

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Title	CURRENT MONITORING DEVICES		
Size	PROC114E3		Rev
C			E3
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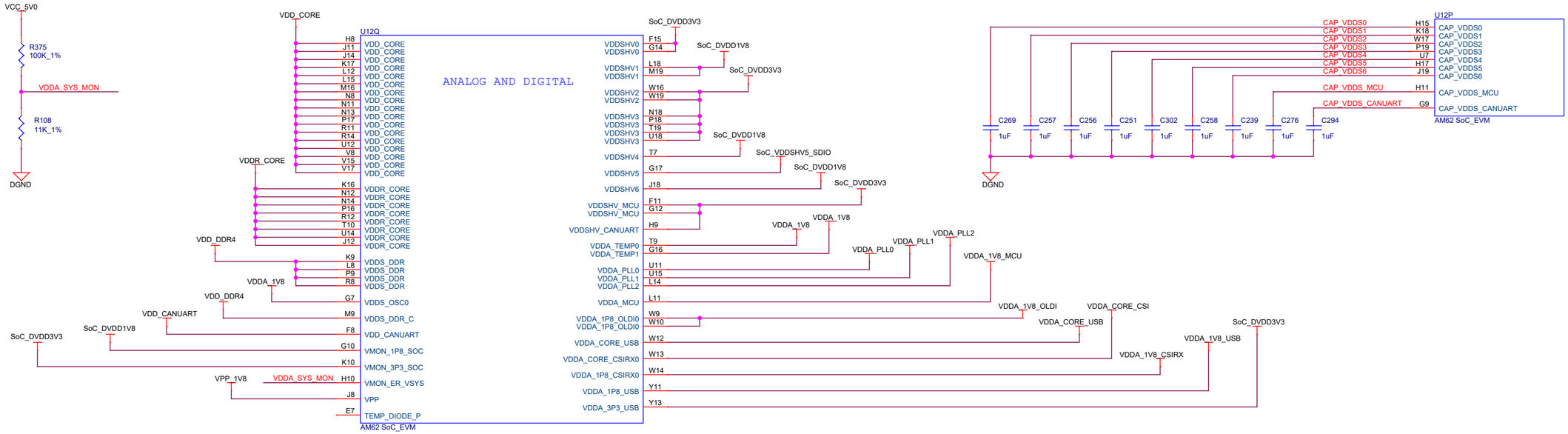
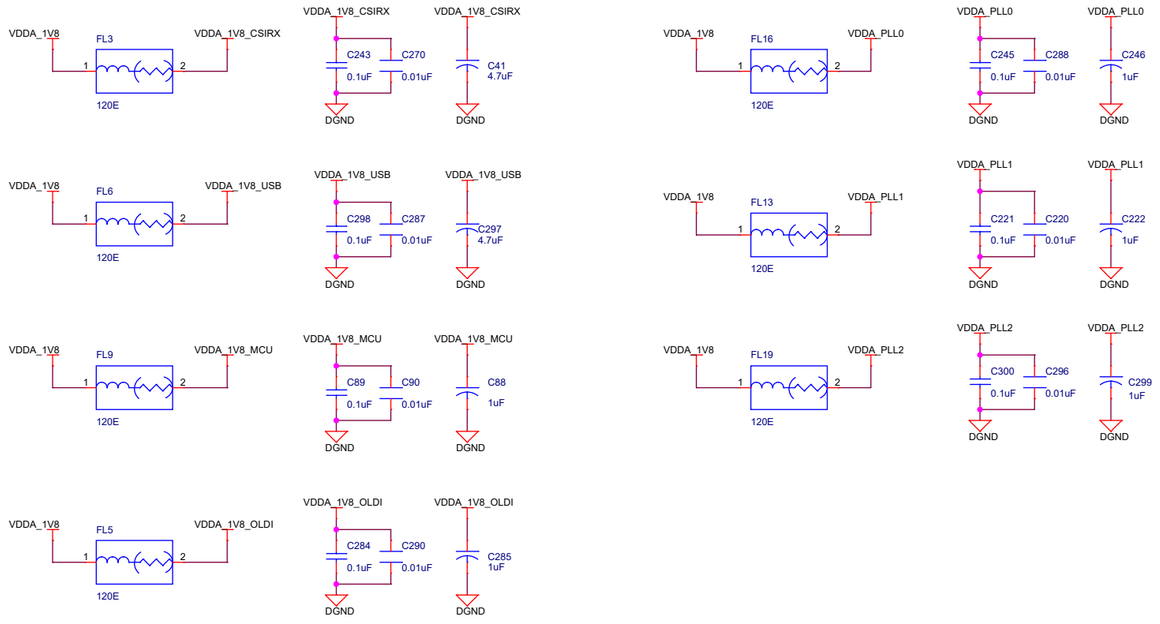
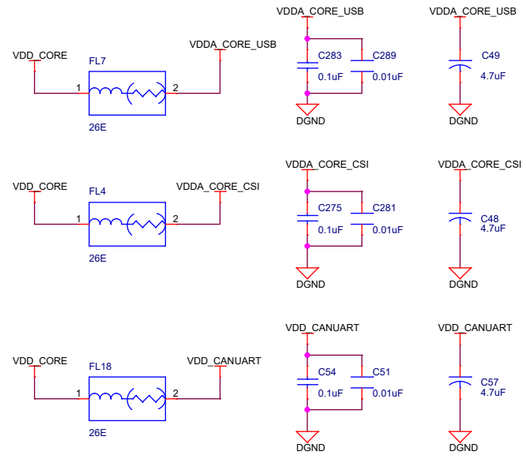
SOC POWER



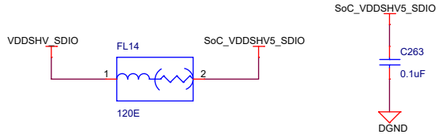
1.8V Analog SUPPLY



CORE SUPPLY



3.3V/1.8V MMC1 SUPPLY



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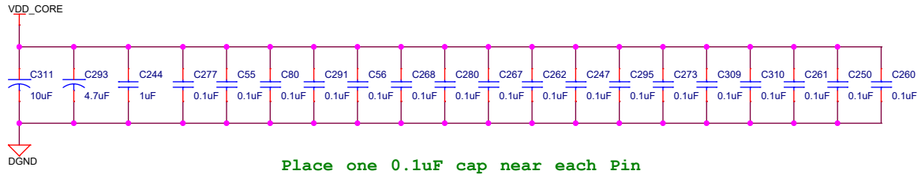
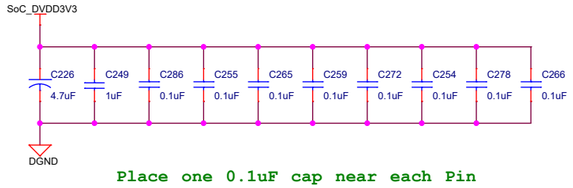
Title	SOC POWER		
Size	Variant Name = PROC114E3		Rev
C			E3
Date:	Wednesday, April 27, 2022	Sheet 14 of 44	

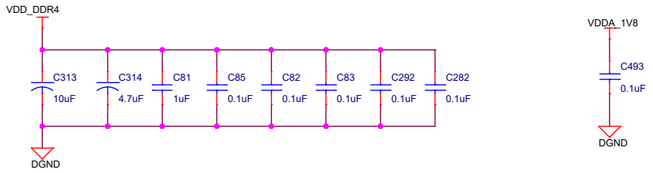
SOC POWER DECAPS



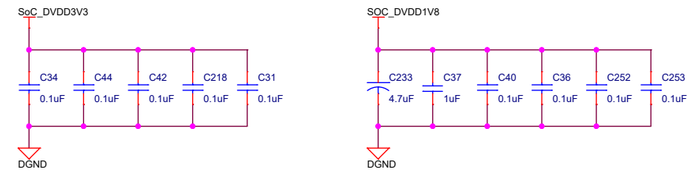
Place one 0.1uF cap near each Pin



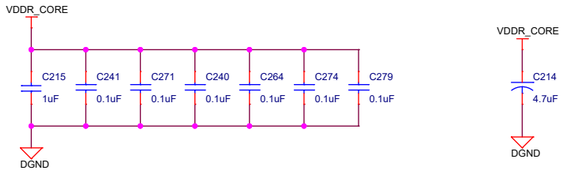
Place one 0.1uF cap near each Pin



Place one 0.1uF cap near each Pin

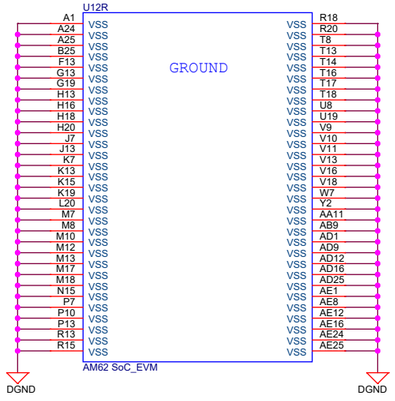


Place one 0.1uF cap near each Pin



Place one 0.1uF cap near each Pin

SOC VSS



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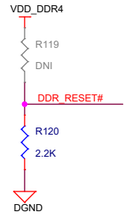
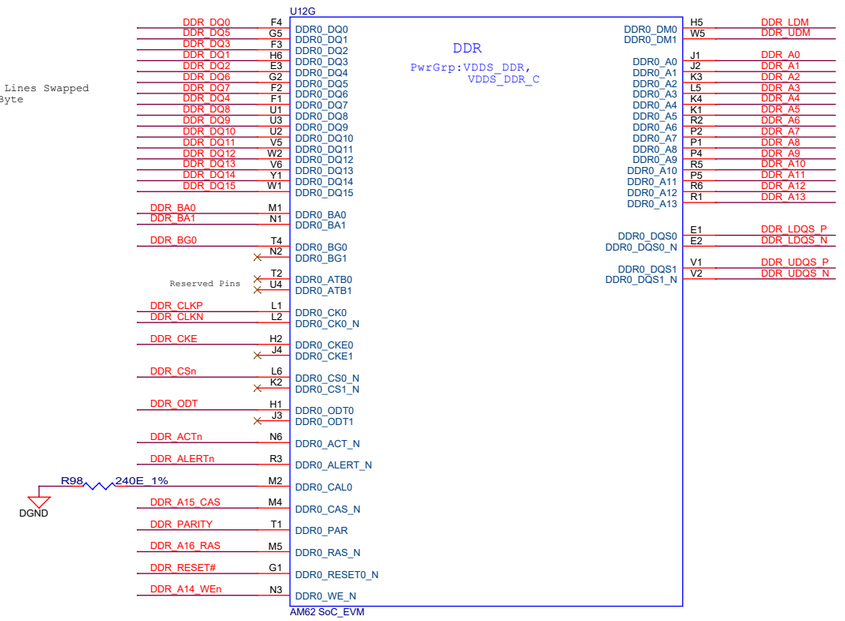
Title	SOC POWER CAPS & SOC VSS	
Size C	Variant Name = PROC114E3	Rev
Date: Wednesday, April 27, 2022	Sheet 15 of 44	

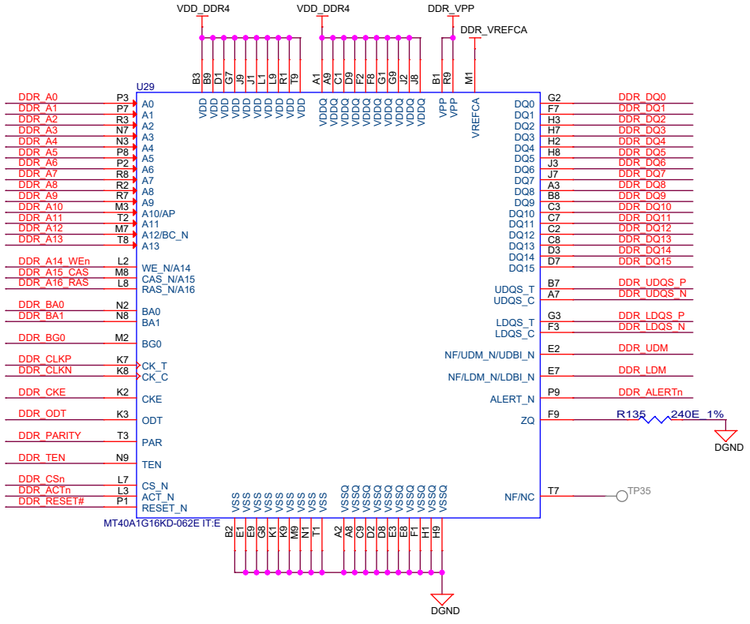
SOC DDR INTERFACE

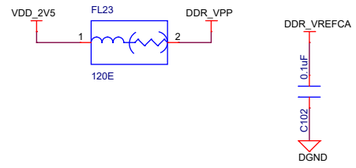
NOTE: DDR DQ Lines Swapped
Within Data Byte

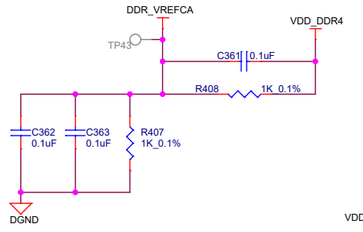


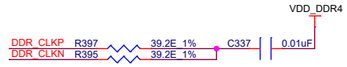


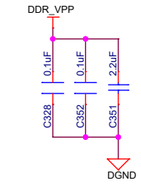
DDR4 DEVICE

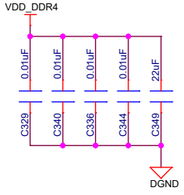


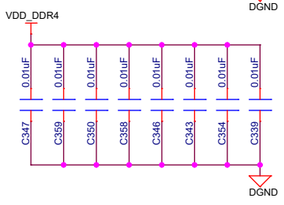


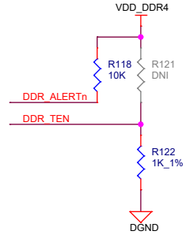


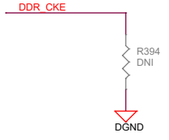












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Title	DDR4 Interface	
Size	Variant Name = PROC114E3	Rev
C		E3
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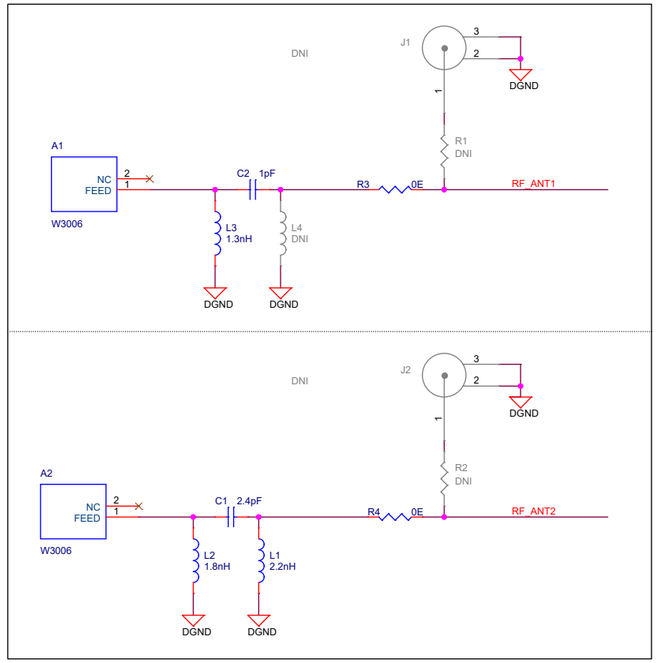
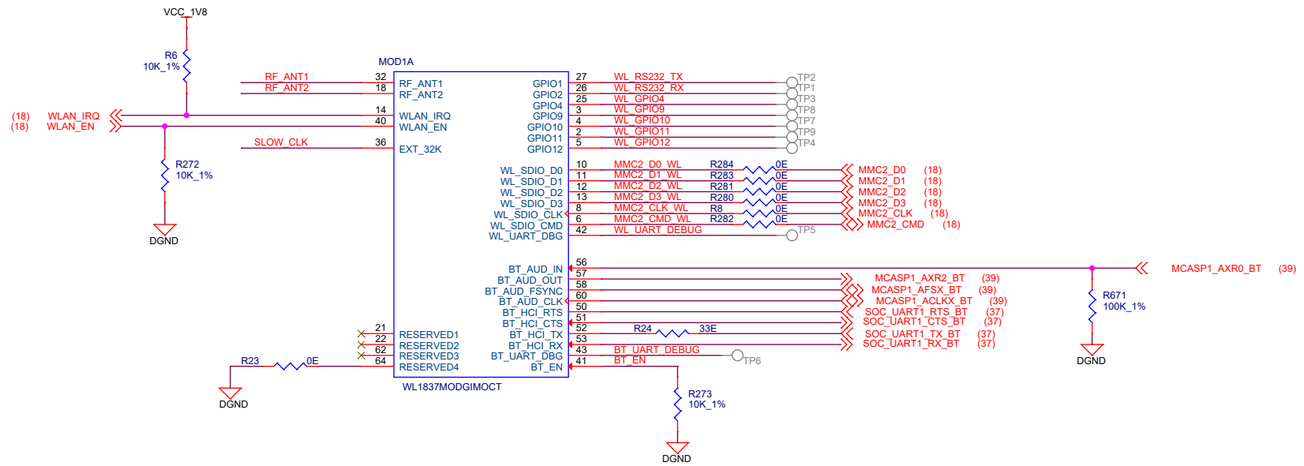
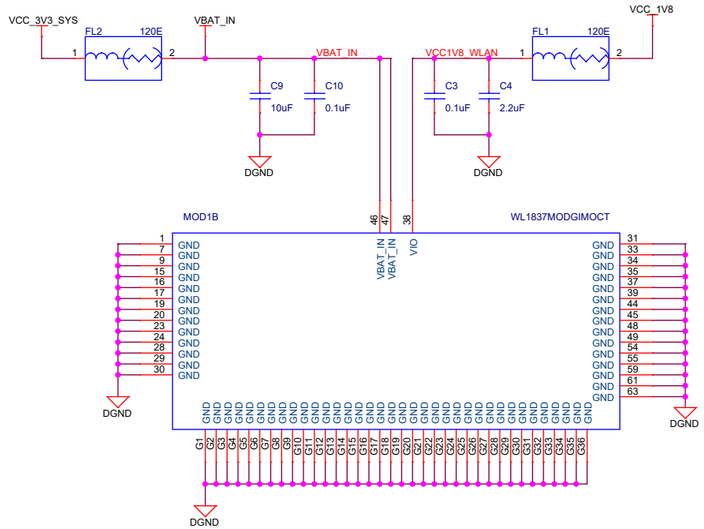
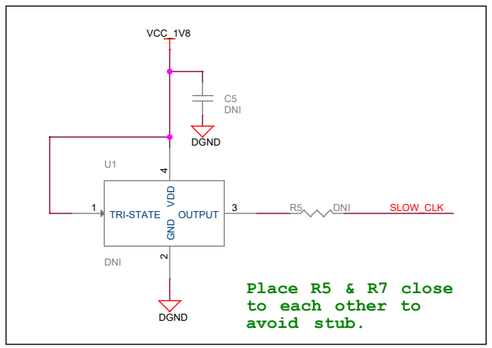
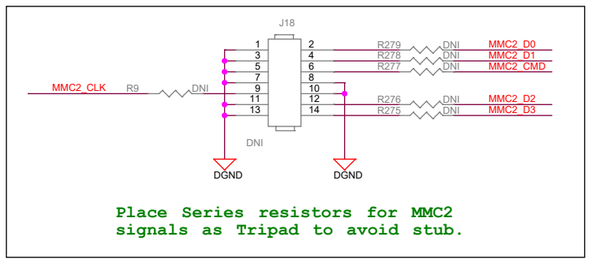
WL1837 MODULE

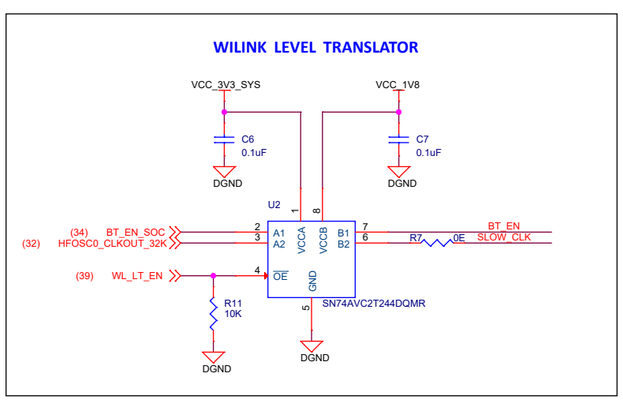


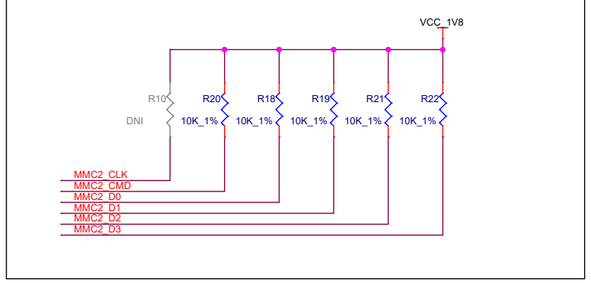












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Title	WL1837 MODULE		
Size	PROC114E3		Rev
C			E3
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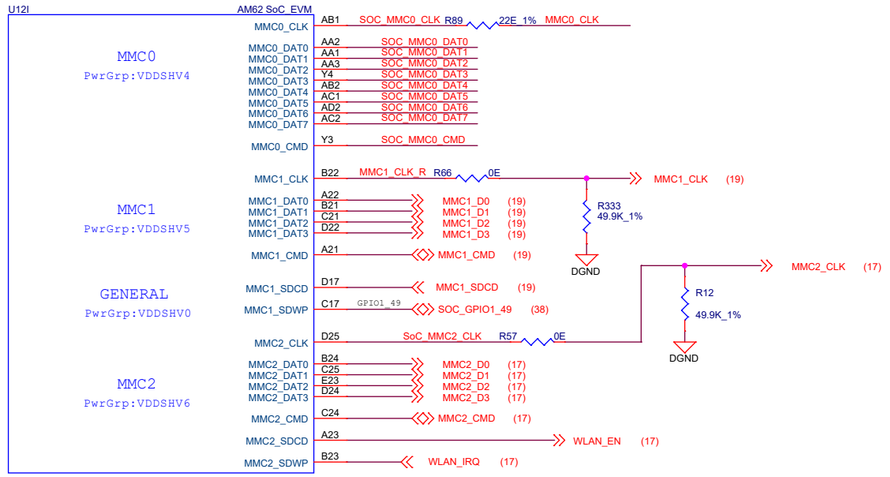
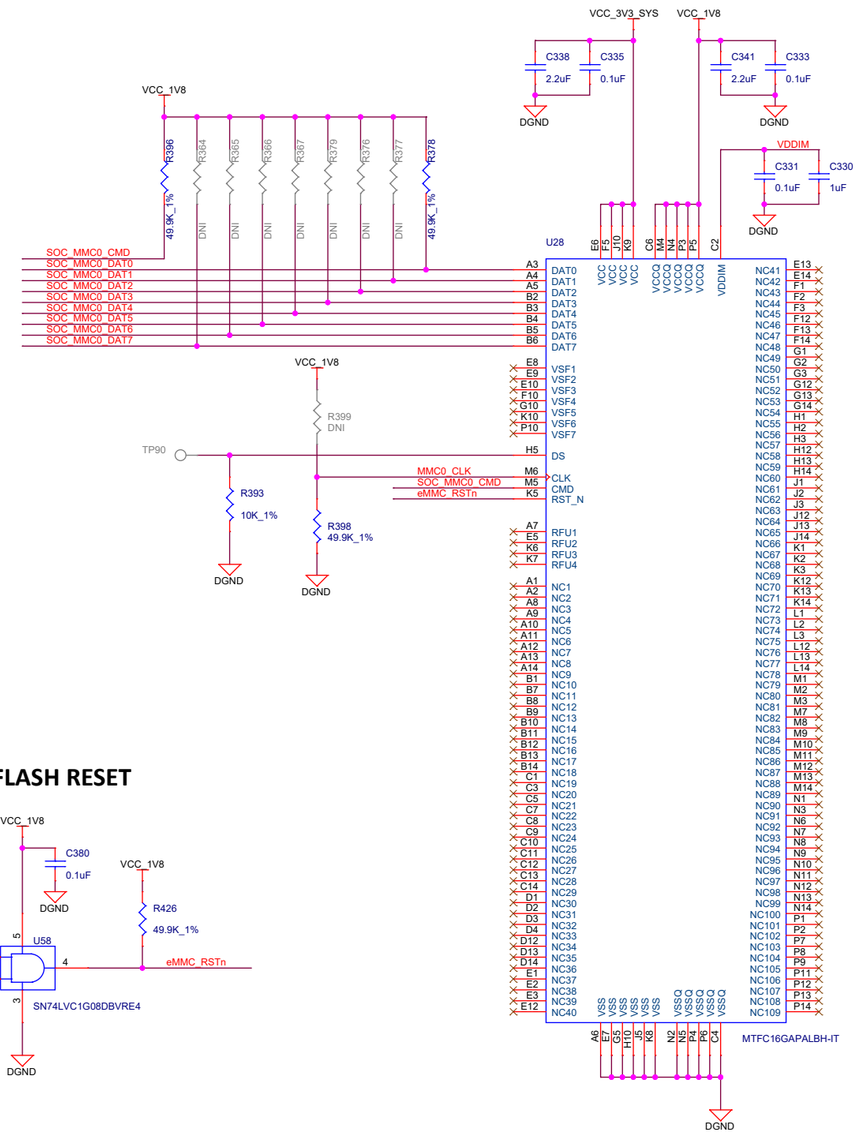
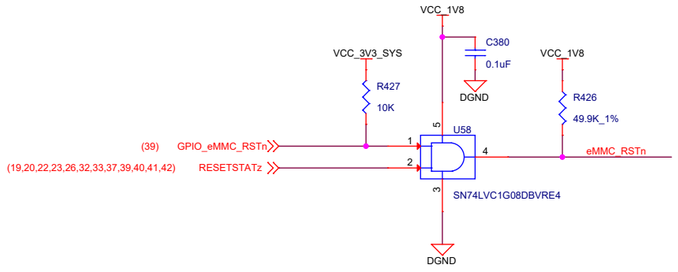
SOC - MMC Interface



eMMC FLASH



eMMC FLASH RESET



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Title	eMMC FLASH INTERFACE	
Size	PROC114E3	Rev
C		E3
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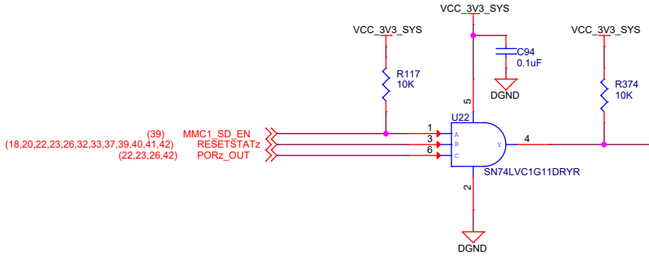
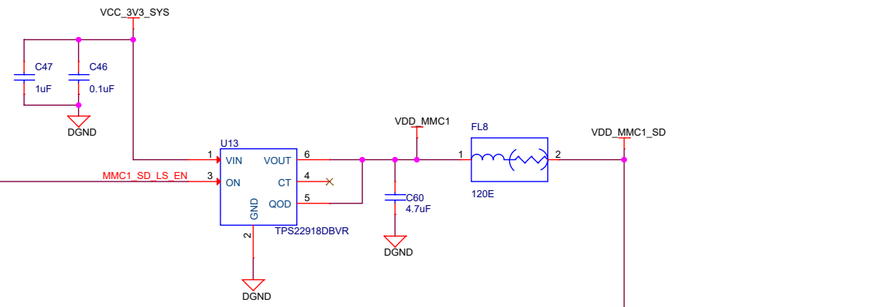
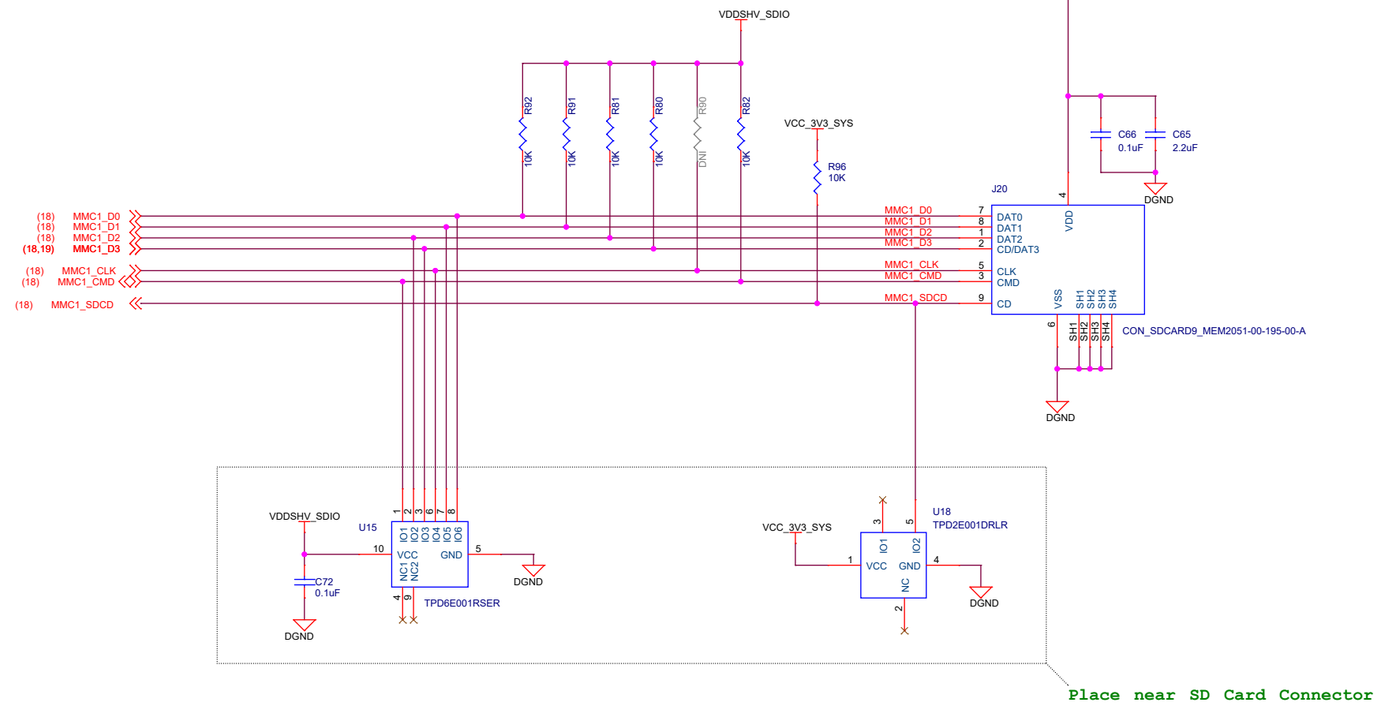
SD CARD INTERFACE

SD CARD RESET



LOAD SWITCH





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Title	SD CARD INTERFACE		
Size	PROC114E3		Rev
C			E3
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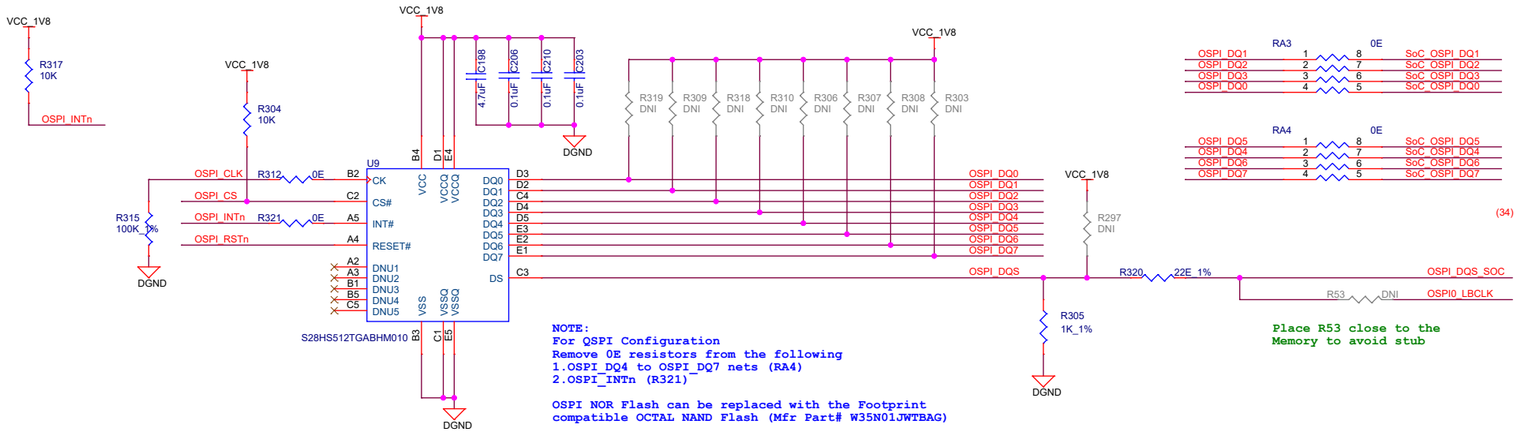
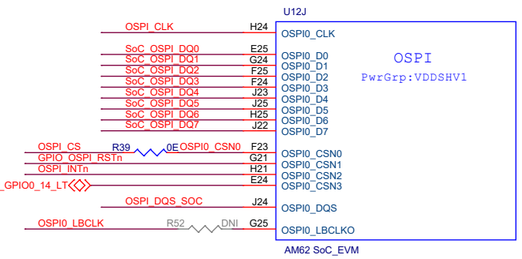
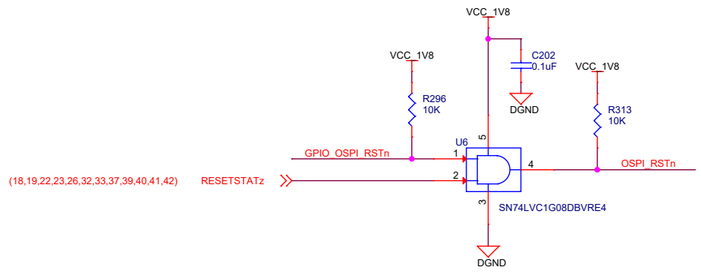
OSPI FLASH



SOC OSPI INTERFACE



OSPI FLASH RESET



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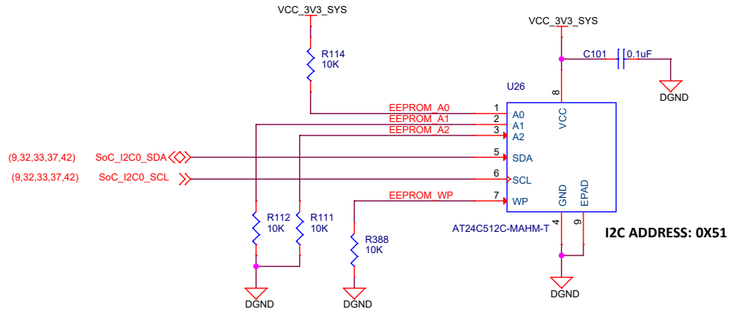
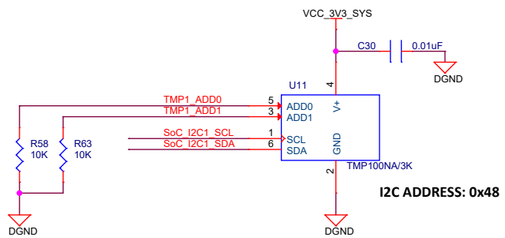
Title	OSPI INTERFACE	
Size: C	PROC114E3	Rev: E3
Date: Wednesday, April 27, 2022	Sheet 20 of 44	

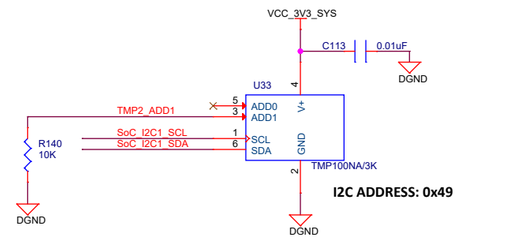
BOARD ID EEPROM



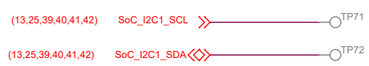
TEMPERATURE SENSORS



CAD NOTE: PLACE TEMP SENSOR U11 CLOSE TO SoC



CAD NOTE: PLACE TEMP SENSOR U33 CLOSE TO DDR4



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Title	BOARD ID EEPROM & TEMPERATURE SENSORS	
Size: C	PROC114E3	Rev: E3
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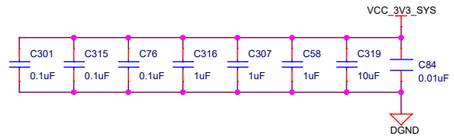
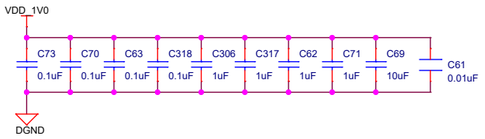
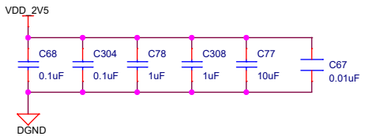
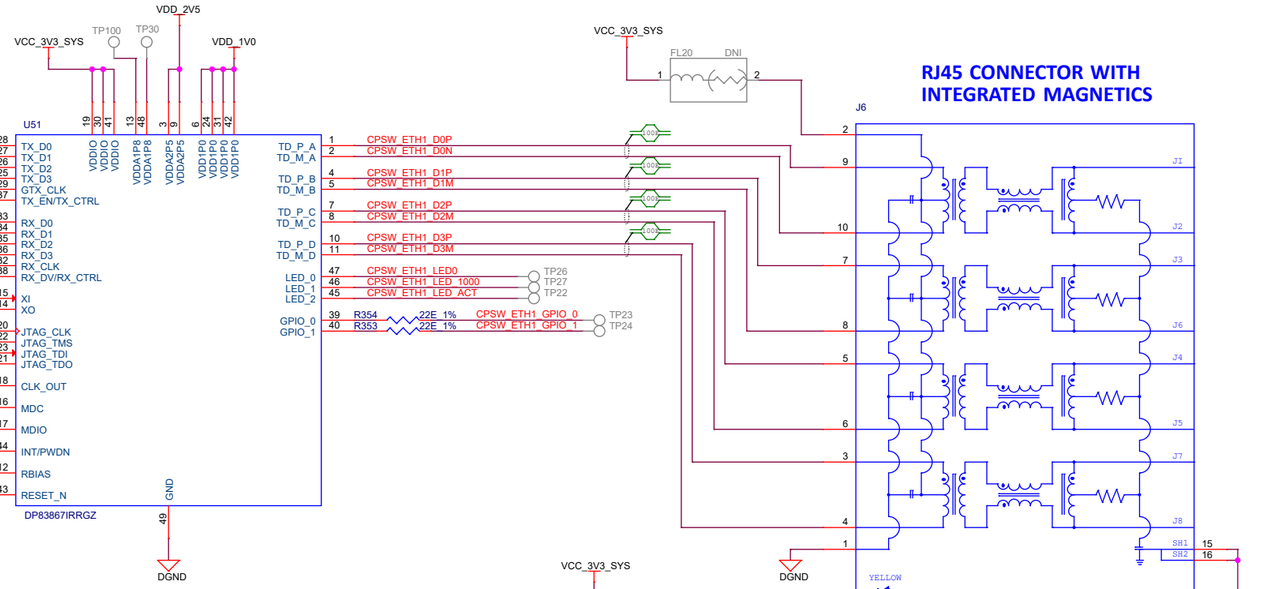
CPSW RGMII 1 - PHY

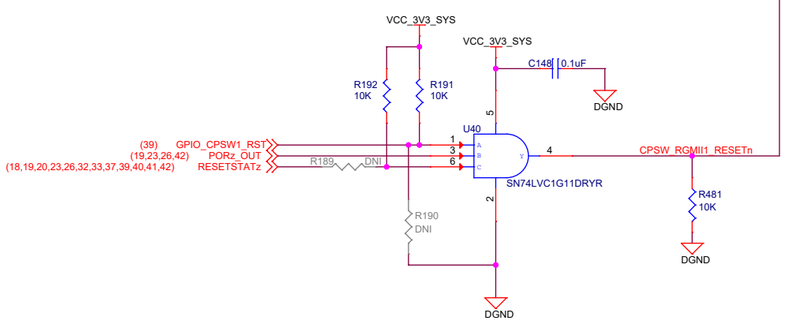


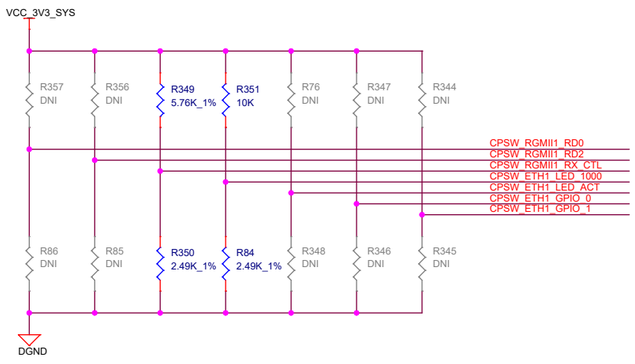


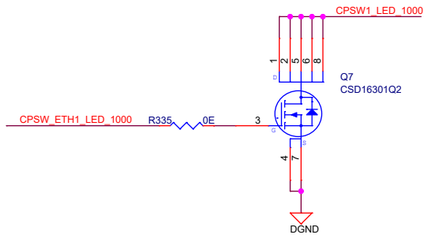


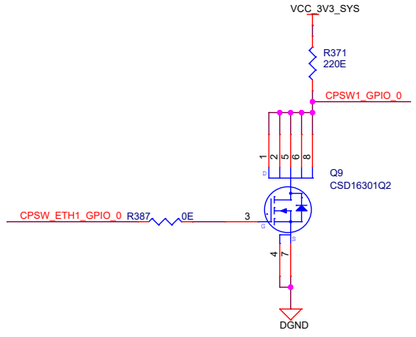


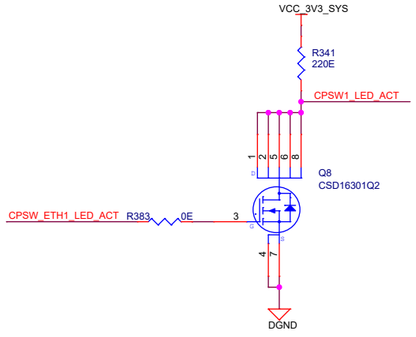
Silk: CPSW PHY-1











PHY ADDRESS = 00000
Auto-negotiation Enabled
10/100/1000 advertised, Auto-MDI-X
Tx Clock Skew = 2ns
Rx Clock Skew = 2ns

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Title	CPSW RGMII_1 ETHERNET PHY	
Size C	PROC114E3	Rev E3
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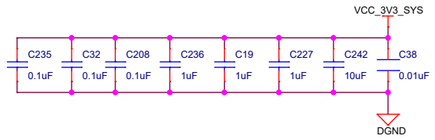
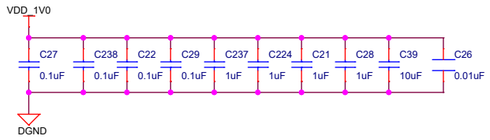
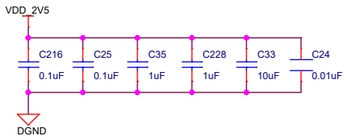
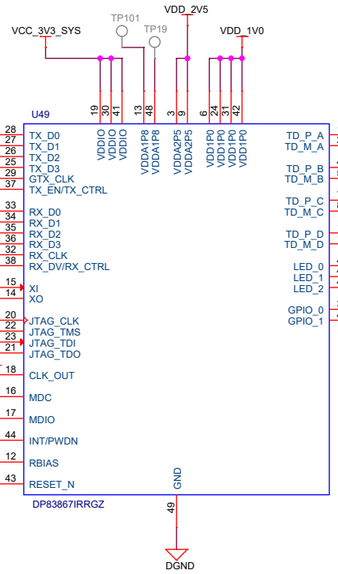
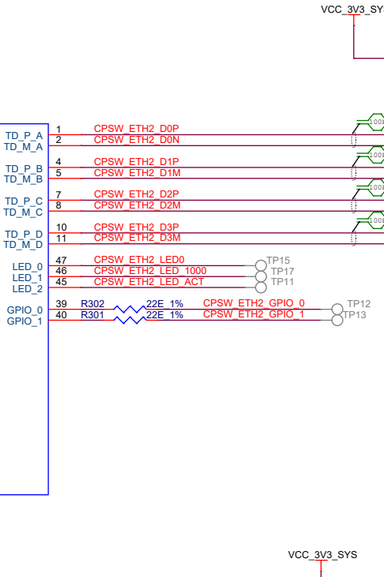
CPSW RGMII 2 - PHY

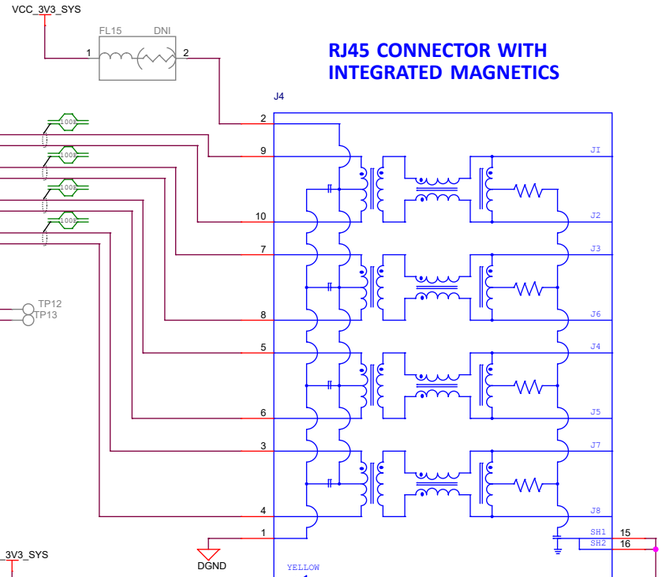


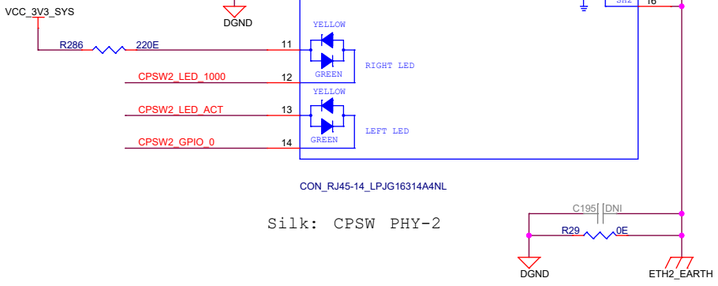


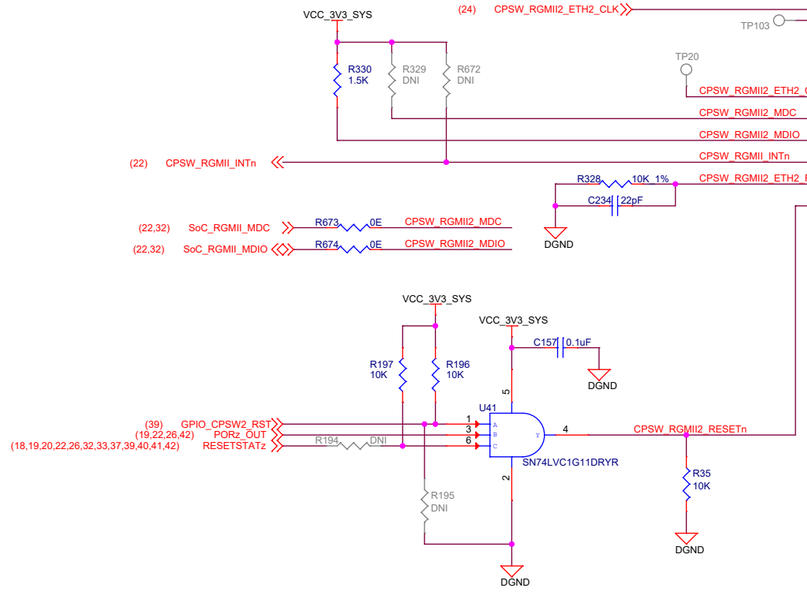


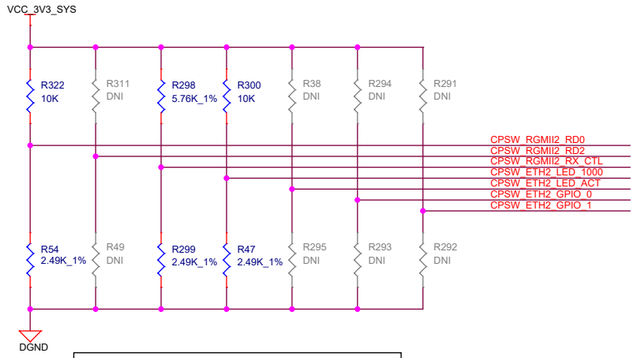




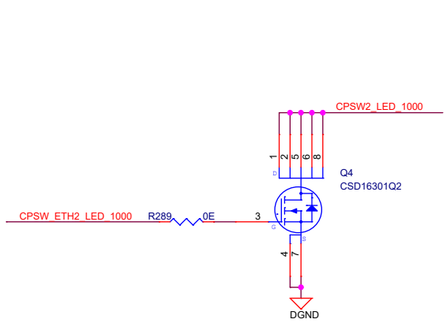


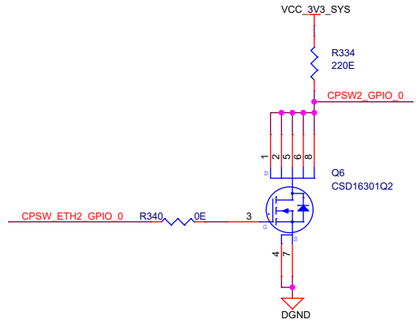


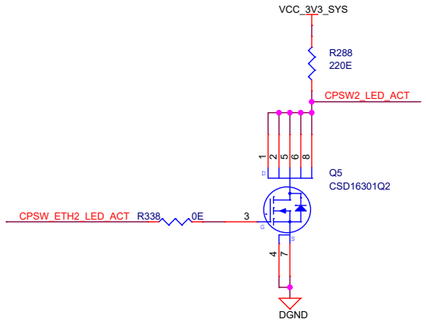




PHY ADDRESS = 00001
Auto-negotiation Enabled
10/100/1000 advertised, Auto-MDI-X
Tx Clock Skew = 2ns
Rx Clock Skew = 2ns







Designed for TI by Mistral Solutions Pvt Ltd

Title	CPSW RGMII_2 ETHERNET PHY		
Size	PROC114E3		Rev
C			E3
Date:	Wednesday, April 27, 2022	Sheet 23 of 44	

AM62 SoC_EVM

U12M

RGMII1
PwrGrp:VDDSHV2

RGMII1_RD0	AB17	CPSW_RGMII1_RD0	(22)
RGMII1_RD1	AC17	CPSW_RGMII1_RD1	(22)
RGMII1_RD2	AB16	CPSW_RGMII1_RD2	(22)
RGMII1_RD3	AA15	CPSW_RGMII1_RD3	(22)
RGMII1_RXC	AD17	CPSW_RGMII1_RXC	(22)
RGMII1_RX_CTL	AE17	CPSW_RGMII1_RX_CTL	(22)
RGMII1_TD0	AE20	CPSW_RGMII1_TD0	(22)
RGMII1_TD1	AD20	CPSW_RGMII1_TD1	(22)
RGMII1_TD2	AE18	CPSW_RGMII1_TD2	(22)
RGMII1_TD3	AD18	CPSW_RGMII1_TD3	(22)
RGMII1_TXC	AE19 — R360 0E	CPSW_RGMII1_TXC	(22)
RGMII1_TX_CTL	AD19	CPSW_RGMII1_TX_CTL	(22)

RGMII2
PwrGrp:VDDSHV2

RGMII2_RD0	AE23	CPSW_RGMII2_RD0	(23)
RGMII2_RD1	AB20	CPSW_RGMII2_RD1	(23)
RGMII2_RD2	AC21	CPSW_RGMII2_RD2	(23)
RGMII2_RD3	AE22	CPSW_RGMII2_RD3	(23)
RGMII2_RXC	AD23	CPSW_RGMII2_RXC	(23)
RGMII2_RX_CTL	AD22	CPSW_RGMII2_RX_CTL	(23)
RGMII2_TD0	Y18	CPSW_RGMII2_TD0	(23)
RGMII2_TD1	AA18	CPSW_RGMII2_TD1	(23)
RGMII2_TD2	AD21	CPSW_RGMII2_TD2	(23)
RGMII2_TD3	AC20	CPSW_RGMII2_TD3	(23)
RGMII2_TXC	AE21 — R65 0E	CPSW_RGMII2_TXC	(23)
RGMII2_TX_CTL	AA19	CPSW_RGMII2_TX_CTL	(23)

ETHERNET PHY CLOCK BUFFER

VCC_1V8 — FL22 120E — VCC1V8_CLKBUF

C326 0.1uF, C325 0.1uF, C324 1uF — DGND

VCC1V8_CLKBUF — R380 10K

(42) CLKOUT0 → U52 LMK1C1103PWR: CLKIN, 1G, NC; VDD = VCC1V8_CLKBUF; GND = DGND; Y0, Y1, Y2

R372 22E 1%, R382 22E 1%, R369 22E 1%

SoC_CLKIN (34)
CPSW_RGMII1_ETH1_CLK (22)
CPSW_RGMII2_ETH2_CLK (23)

C322 DNI, R368 DNI — DGND

LED DRIVER

VCC_3V3_SYS

R175 220E	LD9 150040VS73220	GREEN	LED1
R171 220E	LD8 150040RS73220	RED	LED2
R170 220E	LD7 150040VS73220	GREEN	LED3
R168 220E	LD6 150040AS73220	AMBER	LED4
R159 220E	LD2 150040VS73220	GREEN	LED5
R161 220E	LD3 150040RS73220	RED	LED6
R166 220E	LD4 150040VS73220	GREEN	LED7
R167 220E	LD5 150040AS73220	AMBER	LED8

VCC_3V3_SYS — C375 0.1uF — DGND

U38 TPIC2810D: A0, A1, A2, SCL, SDA, G, VCC, GND, DRAIN0–DRAIN7

(32) WKUP_I2C0_SCL
(32) WKUP_I2C0_SDA

R160 10K, R162 10K, R169 10K — DGND

I2C ADDRESS: 0x60

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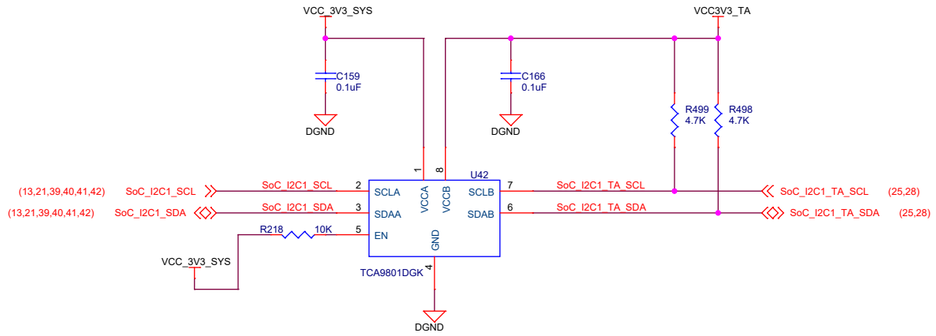
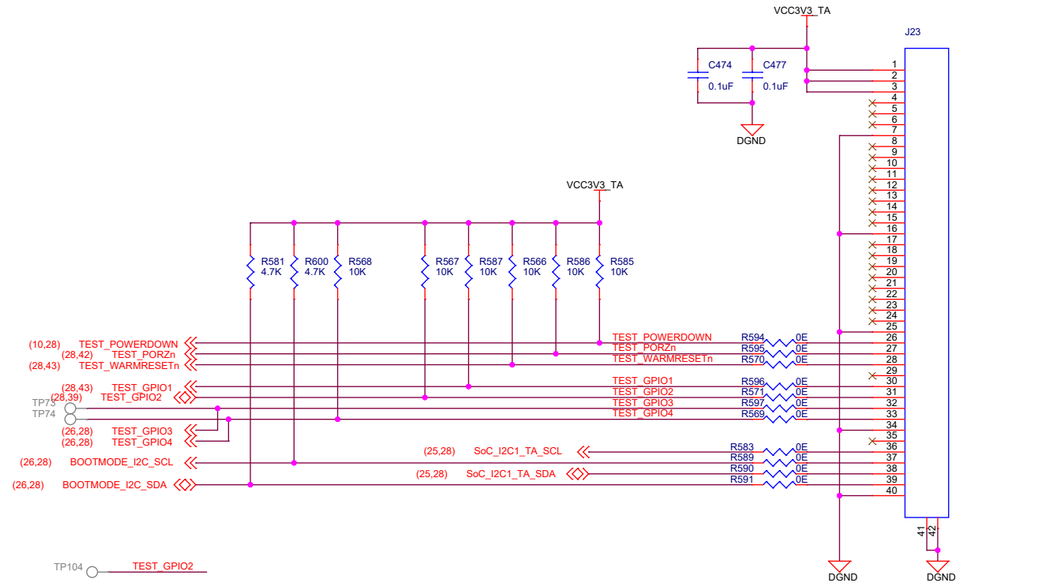
Title	ETHERNET PHY CLOCK BUFFER & LED DRIVER		
Size	C	PROC114E3	Rev E3
Date:	Wednesday, May 04, 2022	Sheet 24 of 44	

I2C BUS BUFFER



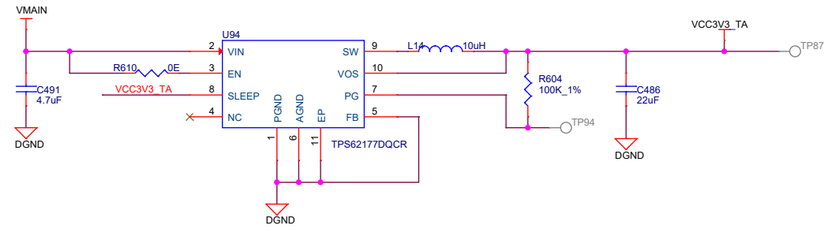
40-PIN TEST AUTOMATION HEADER



Silk: AUTOMATION HDR

TEST AUTOMATION BOARD POWER

VinMin = 4.75V
VinMax = 24V
Vout = 3.3V @ 0.5A



TEST AUTOMATION GPIO MAPPING

SIGNAL NAME	DESCRIPTION	Direction WRT CTRL	Internal/ External PU/PD states
TEST_POWERDOWN	Used to Power down the EVM	OUTPUT	External Pullup
TEST_PORZn	Used to Reset the SoC PORz	OUTPUT	External Pullup
TEST_WARMRESETn	Used to Reset the SoC Warmreset	OUTPUT	External Pullup
TEST_GPIO1	Used to Generate the interrupt on SoC_GPIO1_23 Pin	OUTPUT	External Pullup
TEST_GPIO2	Connected to IO Expander to Communicate with SOC	OUTPUT	External Pullup
TEST_GPIO3	Used to Enable the BOOTMODE Buffer	OUTPUT	External Pullup
TEST_GPIO4	Used to Reset the Bootmode I2C IO Expander	OUTPUT	External Pullup

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Title: TEST AUTOMATION

Size		Rev
C	PROC114E3	E3
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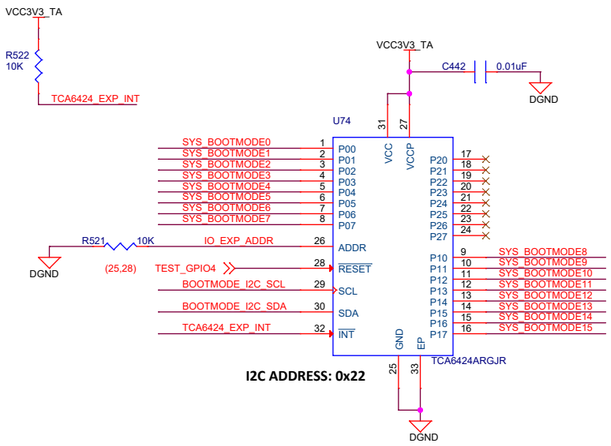
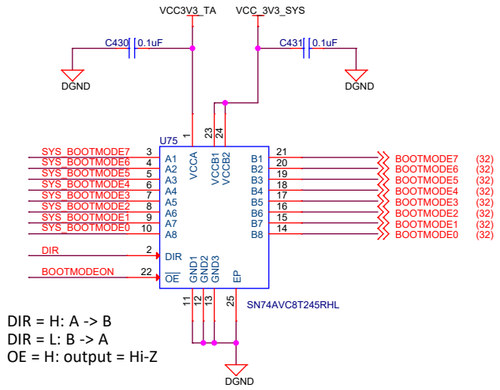
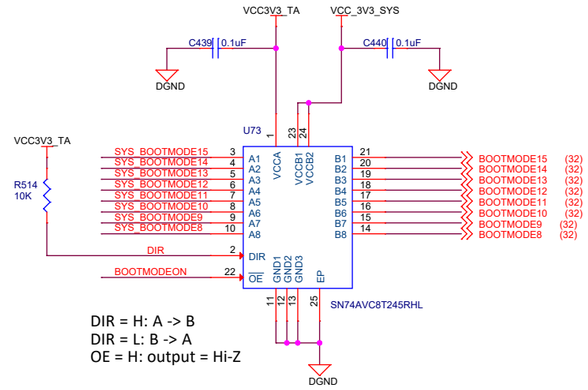
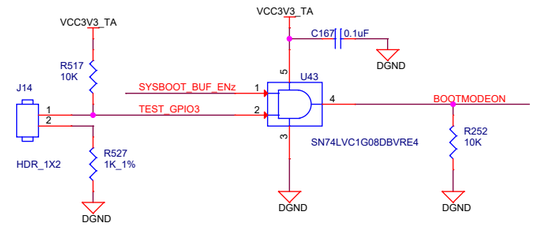
BOOTMODE IO EXPANDER



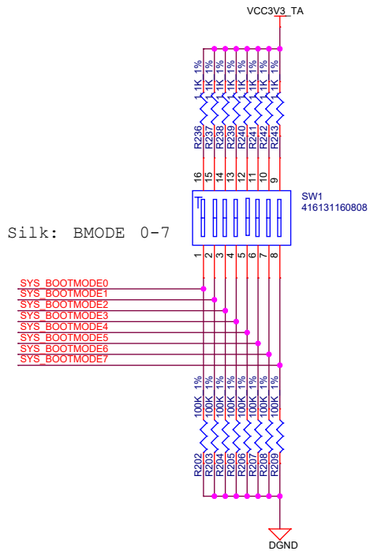
BOOT MODE BUFFERS



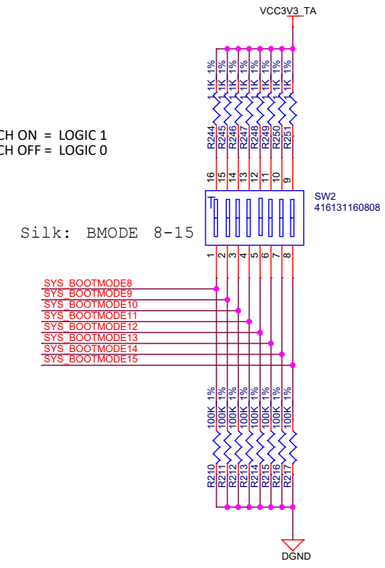




BOOT MODE SWITCHES



SWITCH ON = LOGIC 1
SWITCH OFF = LOGIC 0



BOOT MODES SUPPORTED

1. OSPI
2. MMC1 - SD CARD
3. UART
4. eMMC
5. Ethernet
6. BACKUP BOOT OPTION

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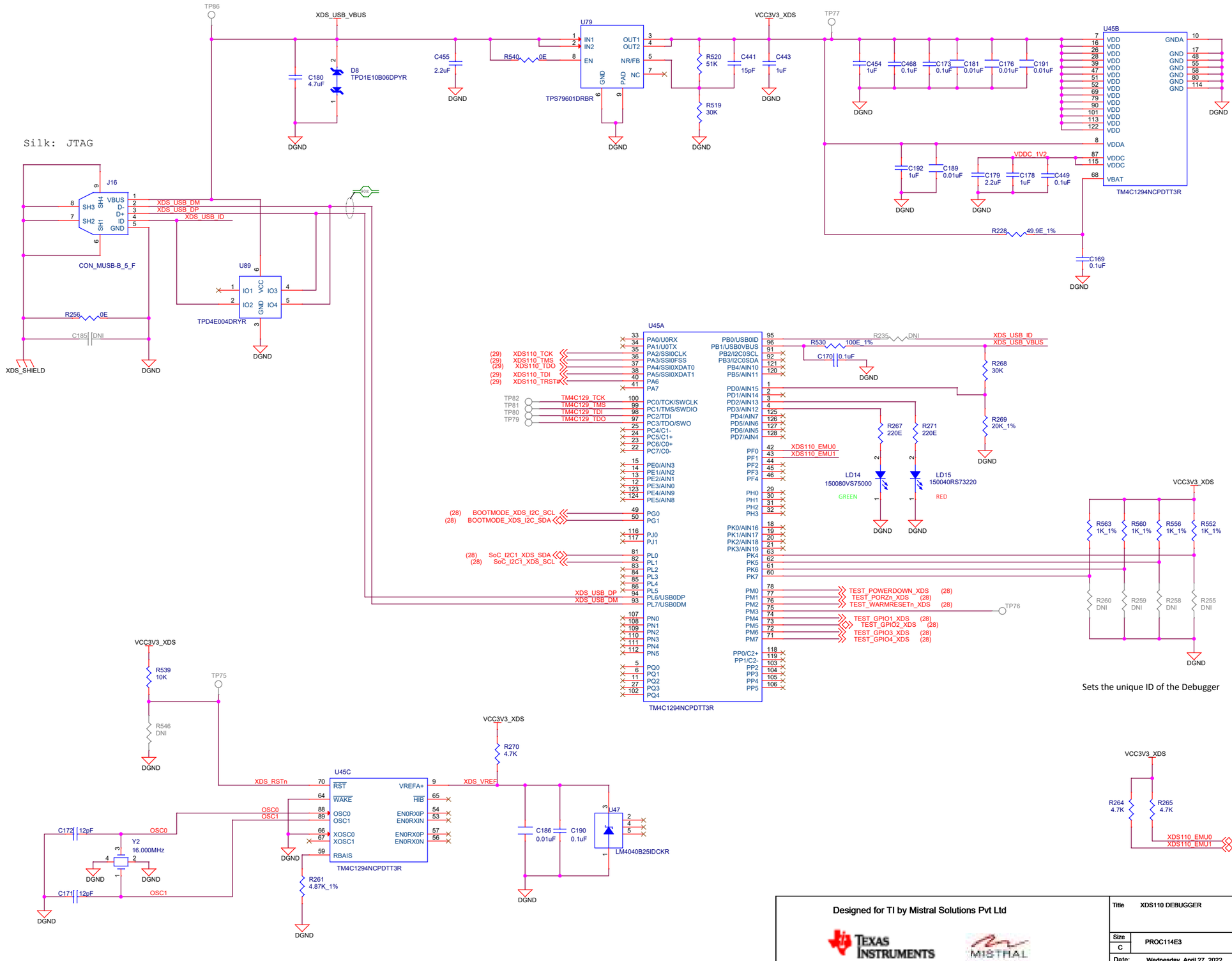
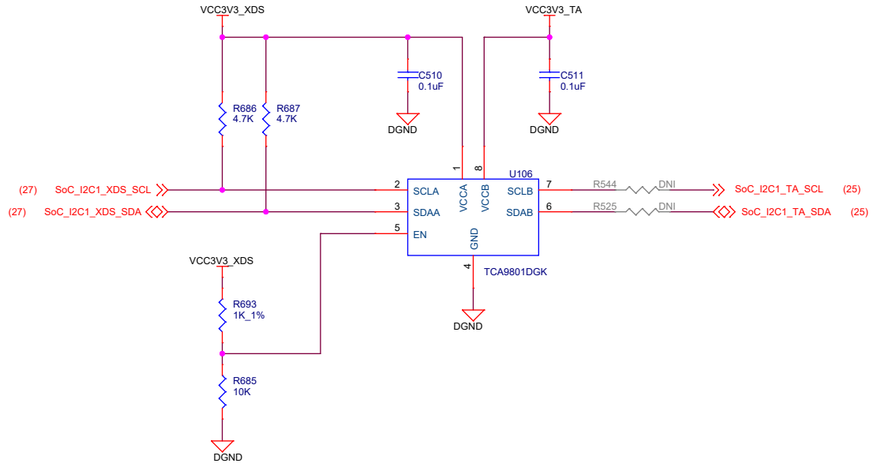
Title	BOOT MODE BUFFER & SWITCHES	
Size C	PROC114E3	Rev E3
Date: Wednesday, April 27, 2022	Sheet 26 of 44	

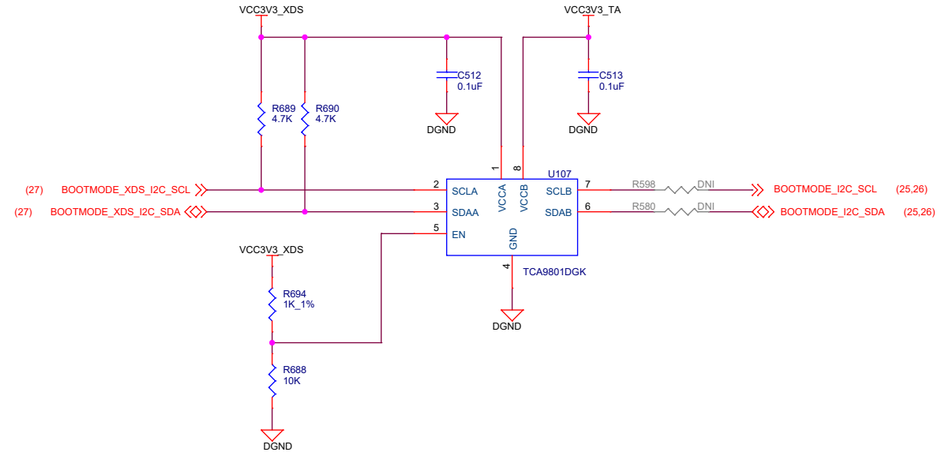
XDS110 DEBUGGER



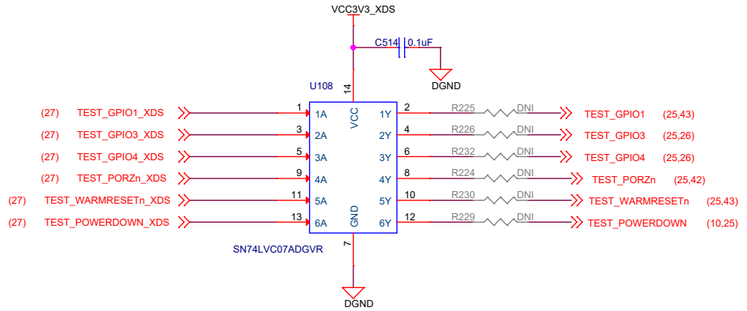
I2C_TA BUS BUFFER

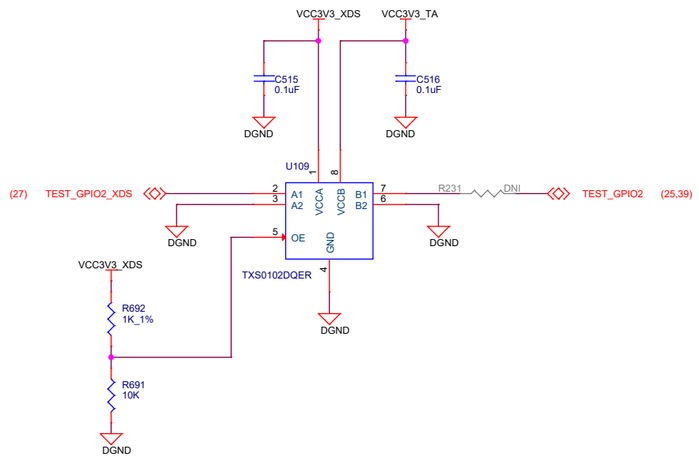


BOOTMODE_I2C_TA BUFFER



ISOLATION BUFFERS FOR TA SIGNALS





Pull ups(R567, R587, R517, R568, R585, R586 & R566) to VCC3V3_TA are present

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Title	AUTOMATION SIGNALS BUFFER	
Size	PROC114E3	Rev
C		E3
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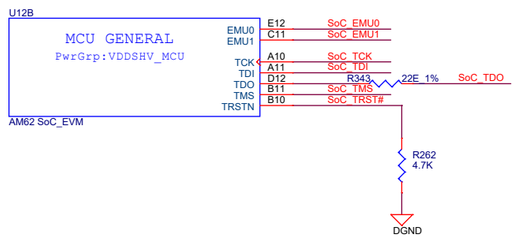
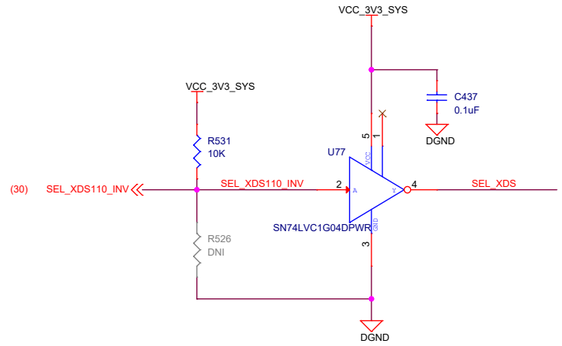
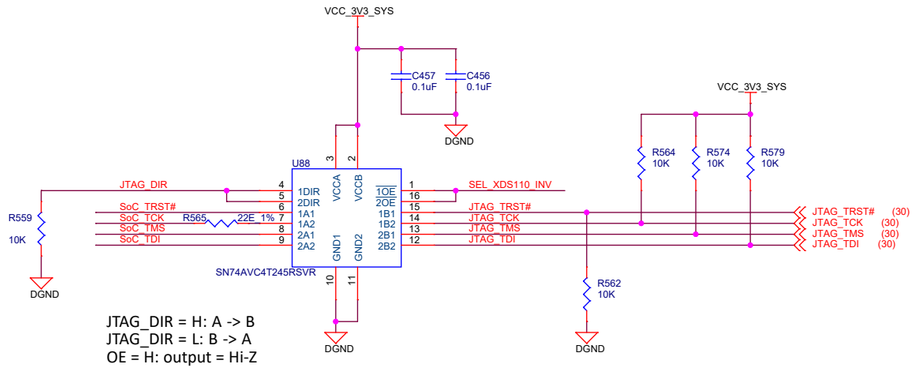
JTAG SOC SECTION



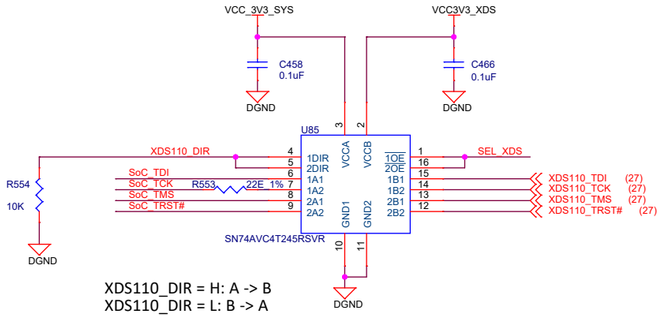


cTI20 JTAG BUFFERS

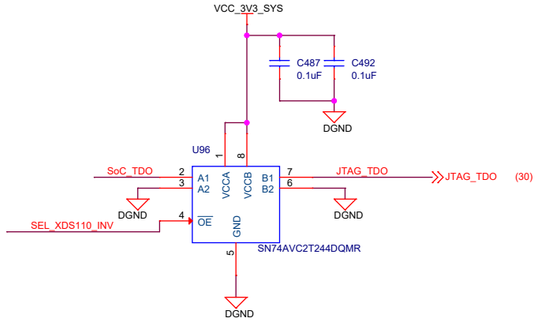


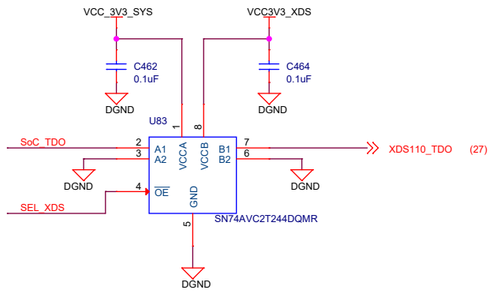
JTAG_DIR = H: A -> B
JTAG_DIR = L: B -> A
OE = H: output = Hi-Z

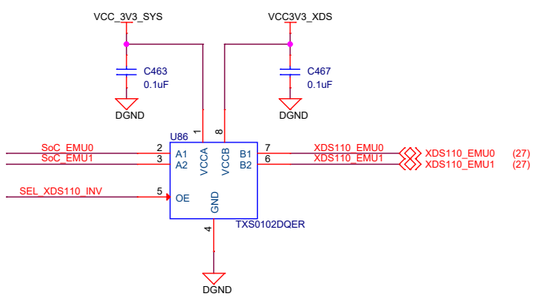
BUFFER XDS110



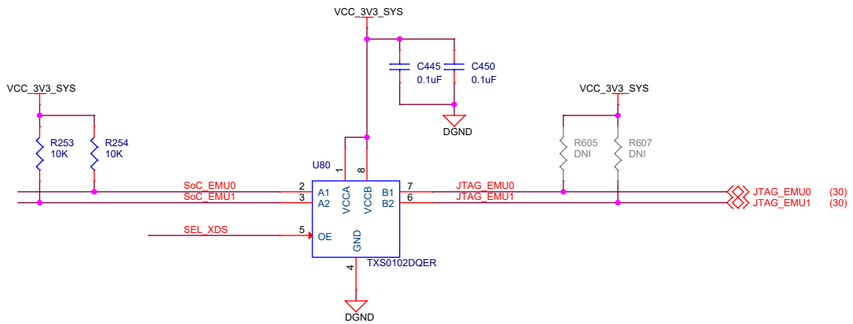
XDS110_DIR = H: A -> B
XDS110_DIR = L: B -> A
OE = H: output = Hi-Z







CAD NOTE: Buffers U88 and U96
need to be placed closer to the
cTI-20pin connector J17 to reduce Stub length of the JTAG signals.



Designed for TI by Mistral Solutions Pvt Ltd

Title	JTAG BUFFER		
Size	PROC114E3		Rev
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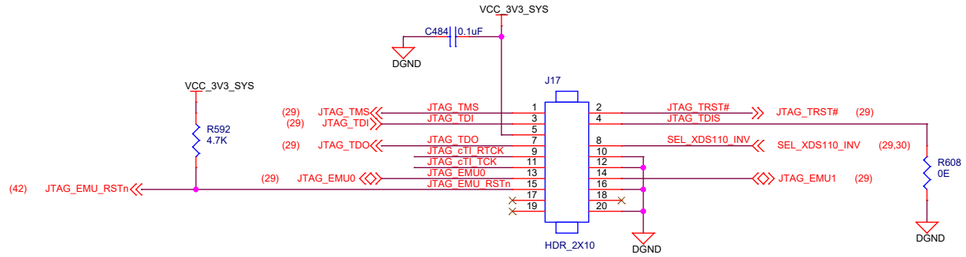
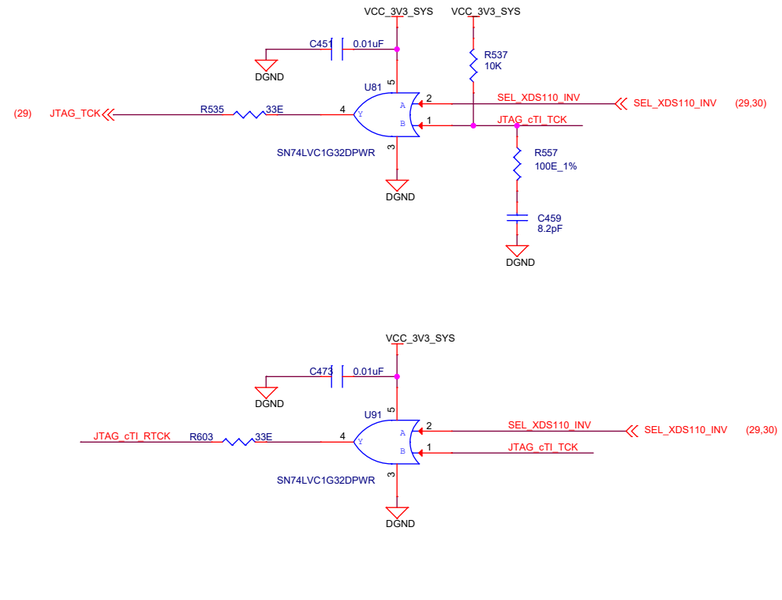
JTAG 20 PIN cTI CONNECTOR



Silk: cTI

JTAG CLOCK BUFFER



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Title	JTAG 20 PIN cTI CONNECTOR		
Size	PROC114E3		Rev
C			E3
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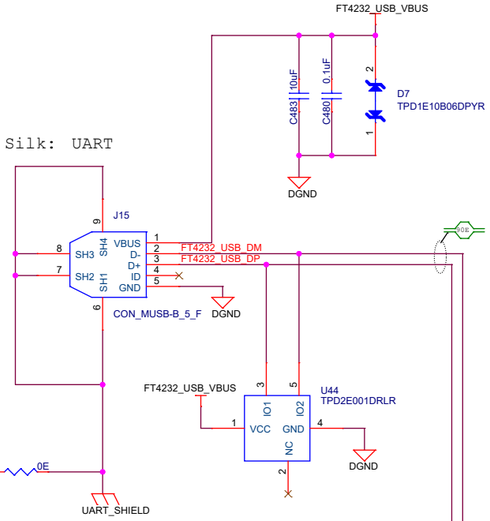
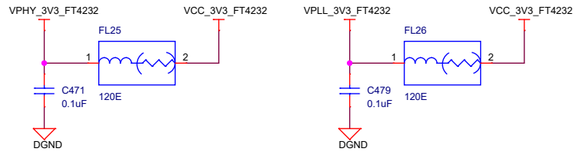
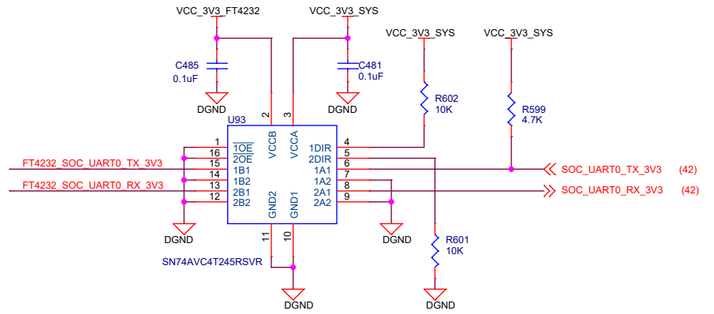
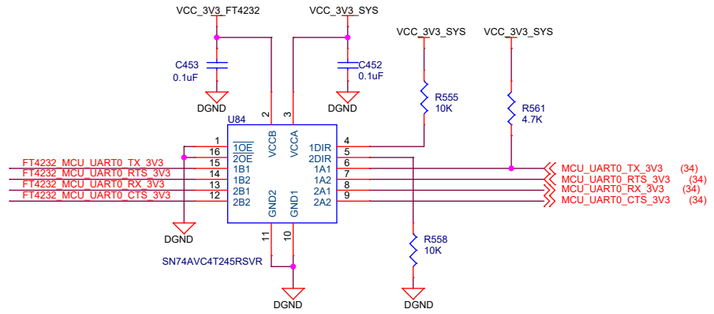
FT4232 UART

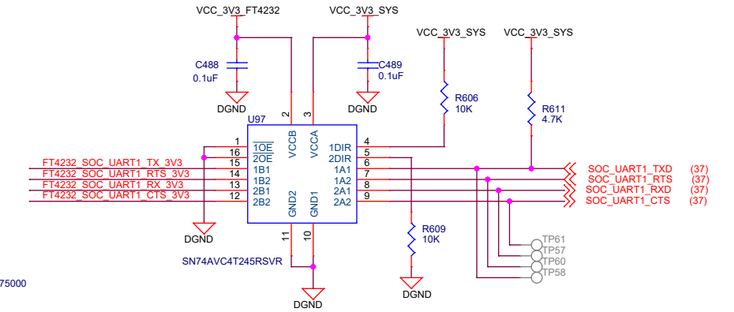
Silk: UART

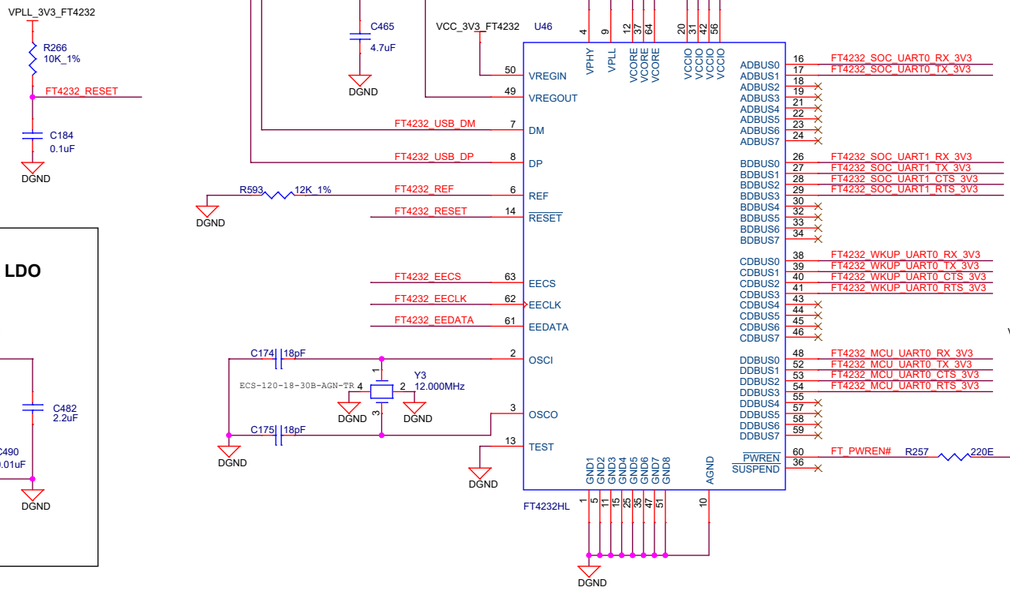




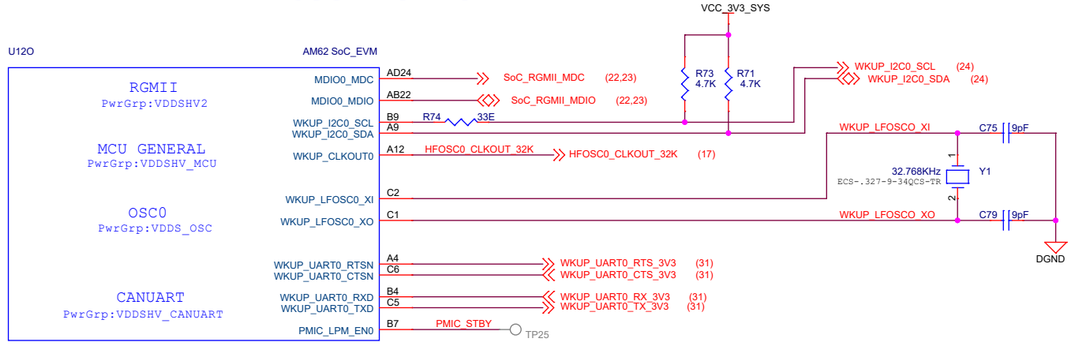




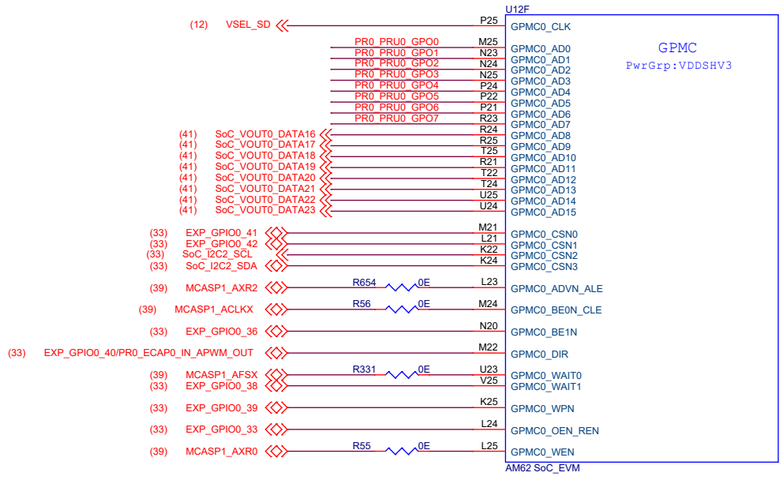




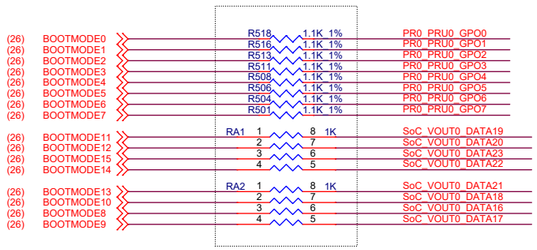
SOC WKUP DOMAIN



SOC GPMC

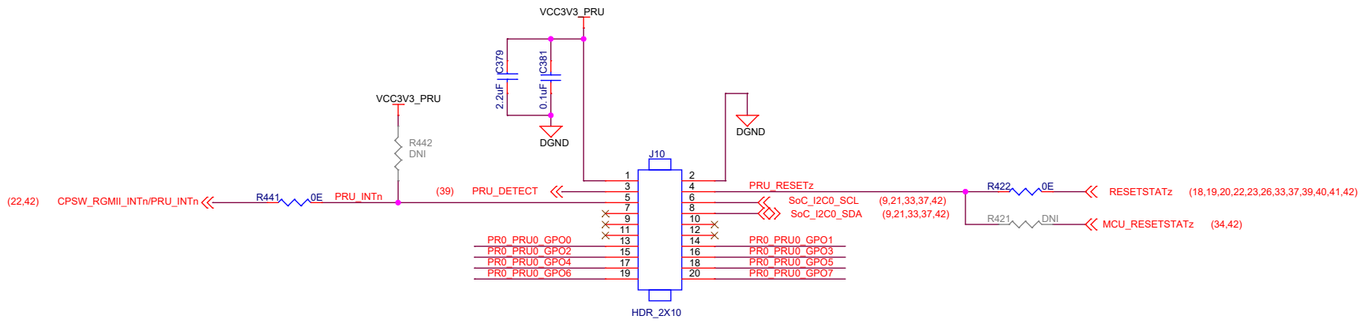


BOOTMODE PINS



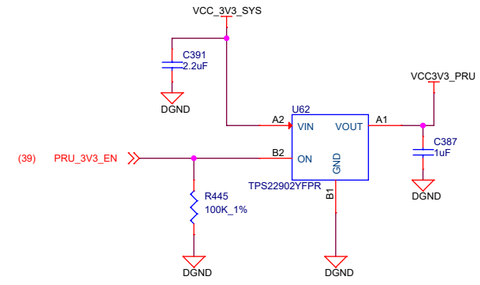
NOTE: Resistors are used to isolate the BOOTMODE control logic after the value is latched

PRU HEADER



Silk: PRU HDR

POWER SWITCH FOR PRU HEADER



3V3 supply of PRU Header is limited to sourcing 500mA max.

NOTE: PRU Header I/O are not fail-safe and shall not be driven when AM62x Starter Kit is not powered.

Designed for TI by Mistral Solutions Pvt Ltd

Title	PRU HEADER	
Size	PROC114E3	Rev E3
C		
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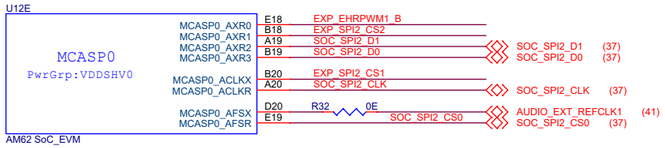
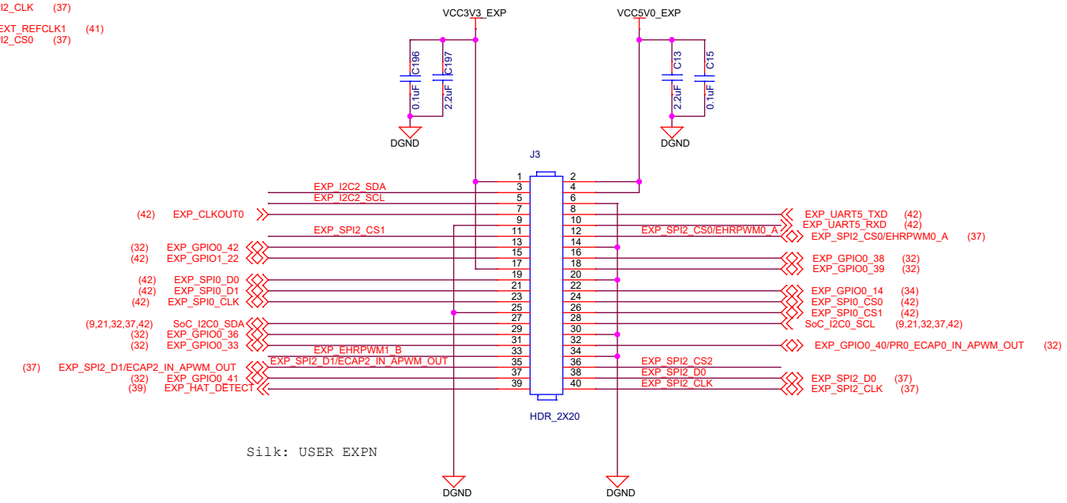
USER EXPANSION CONNECTOR

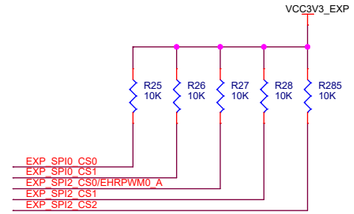


R32(Series damping resistor) should be placed close to SoC

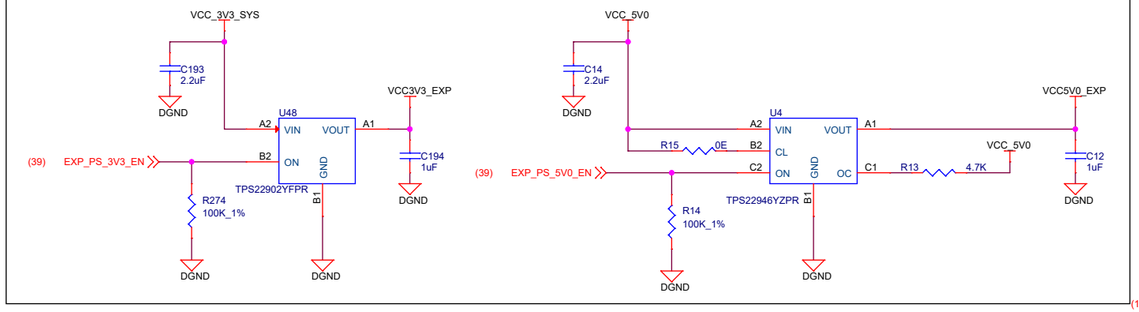


Silk: USER EXPN

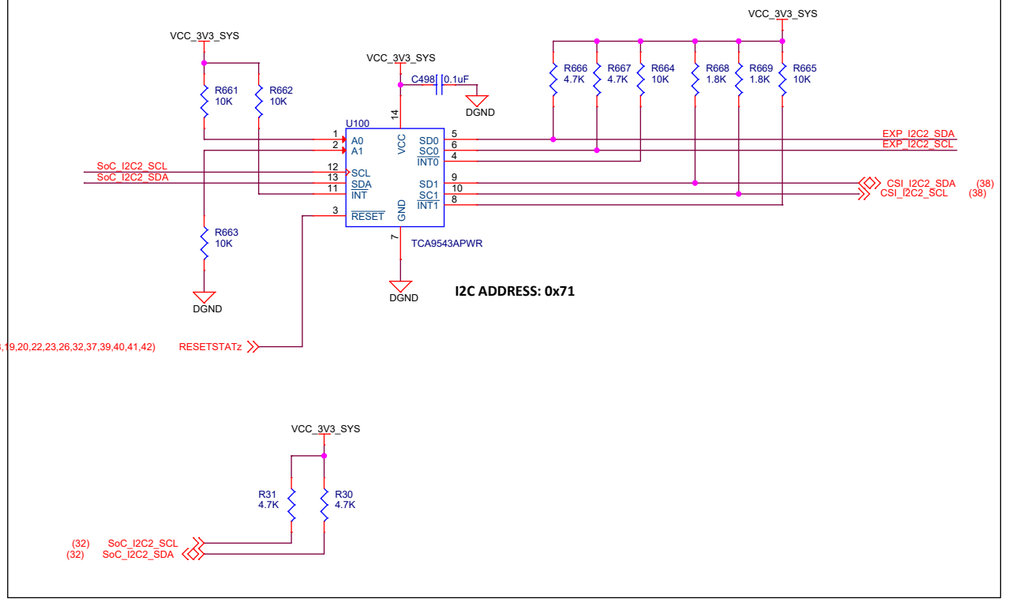
Note: Expansion boards should take care of the null modem connectivity for the UART signals (cross-over of Rx and Tx)



POWER SWITCHES FOR USER EXPANSION CONNECTOR



I2C SWITCH FOR SoC_I2C2



NOTE:

AM62x Starter Kit shall not be powered through the 5V0 or 3V3 pins on the 40-pin User Expansion Connector.

User Expansion Connector I/O are not fail-safe and shall not be driven when AM62x Starter Kit is not powered.

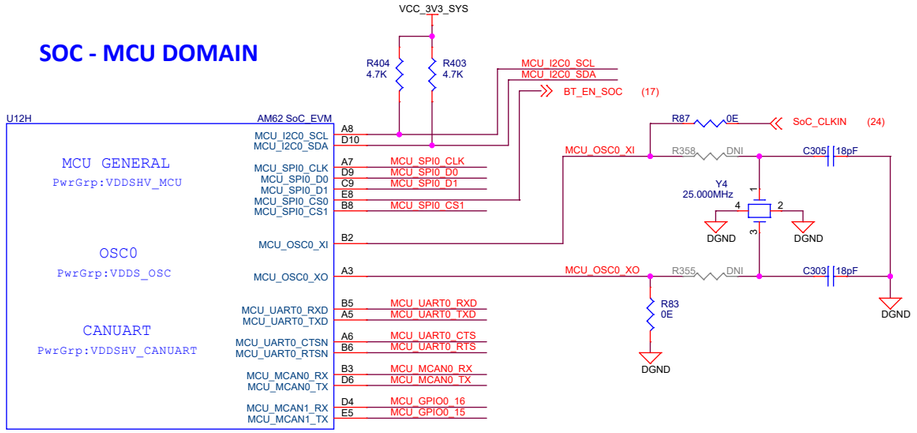
5V supply of User Expansion Connector is limited to sourcing 155mA max.

3V3 supply of User Expansion Connector is limited to sourcing 500mA max.

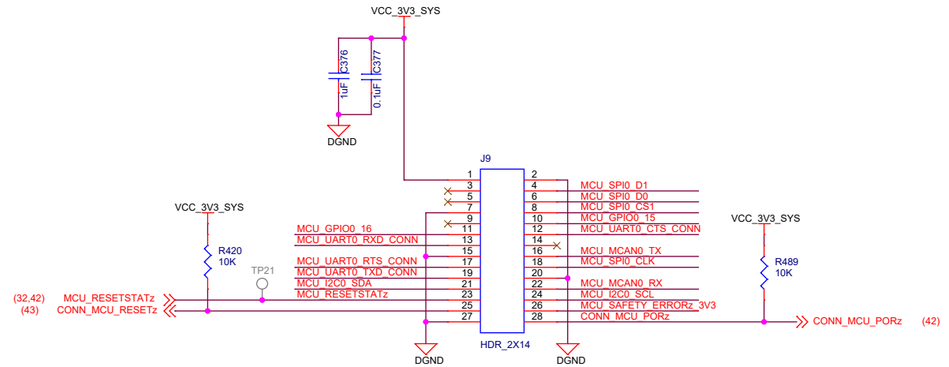
Designed for TI by Mistral Solutions Pvt Ltd

Title	USER EXPANSION CONNECTOR	
Size C	PROC114E3	Rev E3
Date: Wednesday, April 27, 2022	Sheet 33 of 44	

SOC - MCU DOMAIN

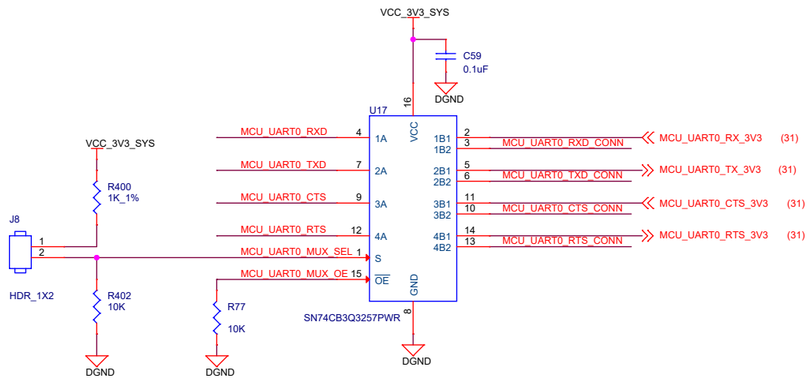


MCU HEADER

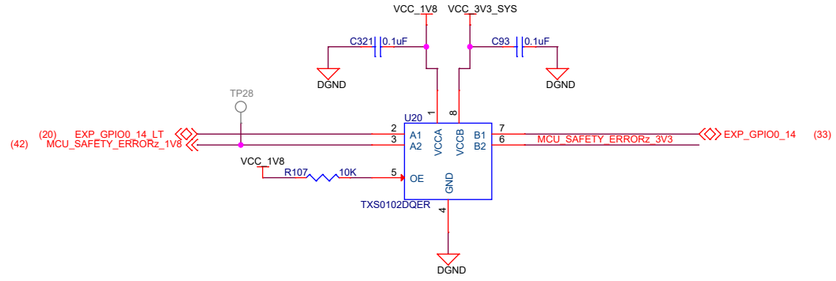


Silk: MCU HDR

MCU_UART0 MUX



OEn	SEL	INPUT/OUTPUT An	
L	L(DEFAULT)	An=nB1	SOC - FT4232
L	H	An=nB2	SOC - MCU HEADER



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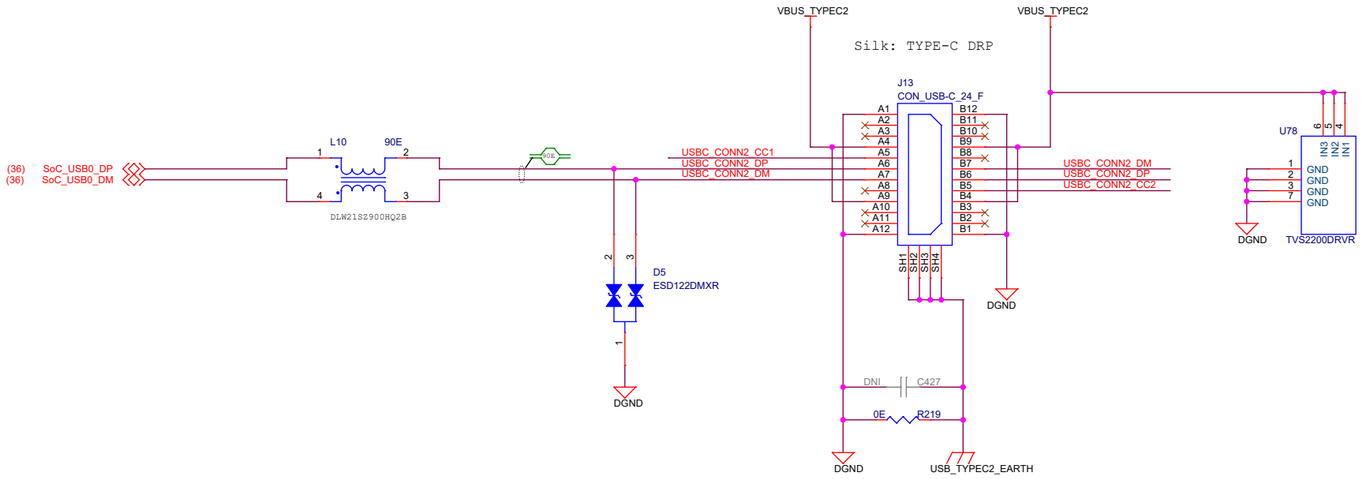
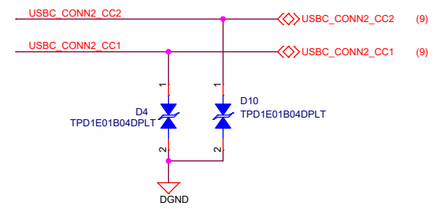
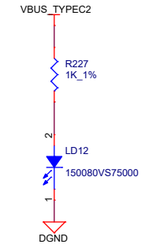
Title	MCU HEADER	
Size C	PROC114E3	Rev E3
Date: Wednesday, April 27, 2022	Sheet 34 of 44	

USB0 TYPE-C DRP

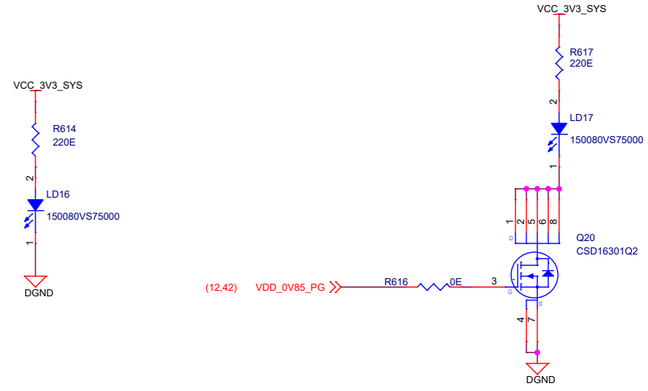




POWER INDICATION LED: VBUS_TYPEC2



POWER RAIL LEDS



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Title	USB0 TYPE-C DRP	
Size: C	PROC114E3	Rev: E3
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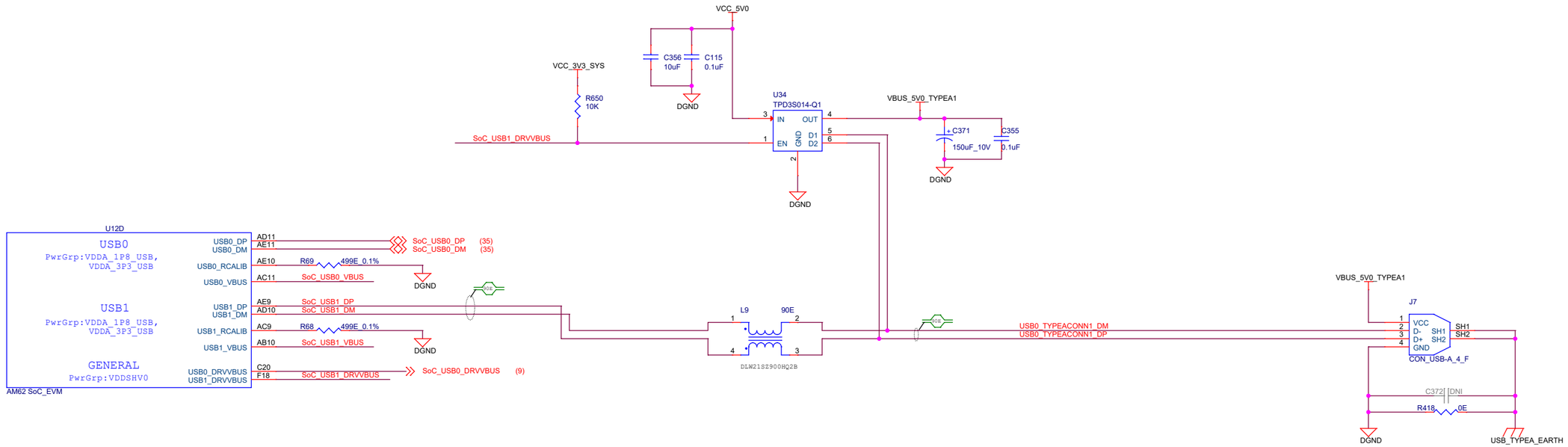
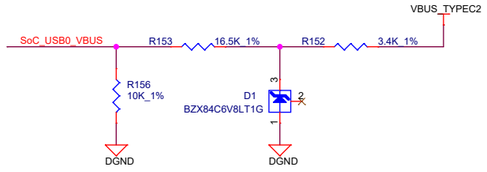
USB1 TYPE-A

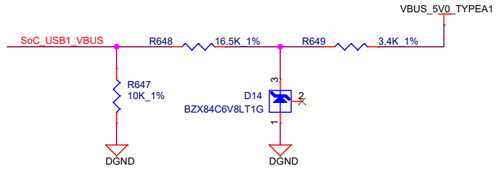


Silk: TYPE-A

Note: Recommended VBUS circuit for USB connector. Supports 5V-30V VBUS



Note: Recommended VBUS circuit for SoC_USB1_VBUS



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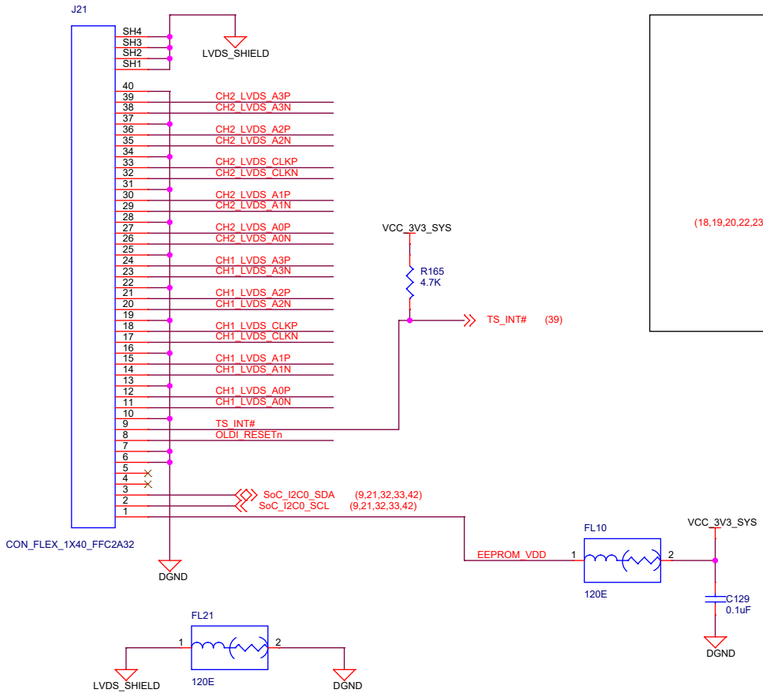
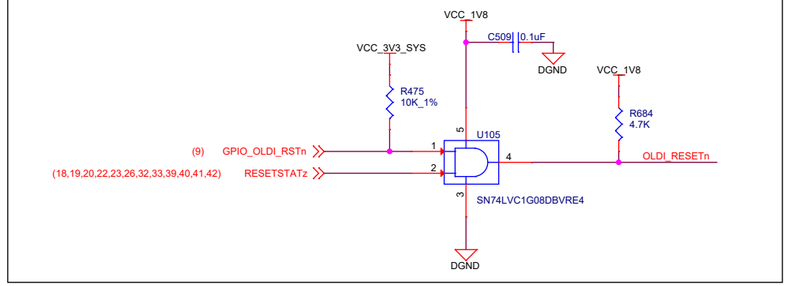
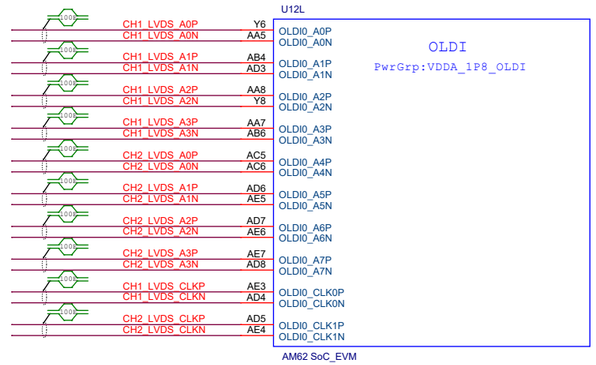
Title	USB1 TYPE-A	
Size: C	PROC114E3	Rev: E3
Date: Wednesday, April 27, 2022	Sheet 36 of 44	

OLDI DISPLAY INTERFACE

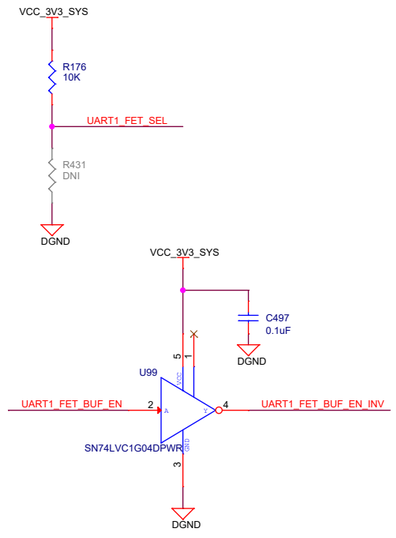


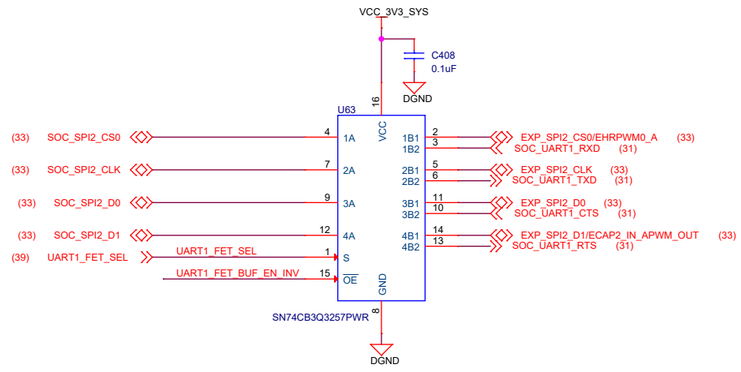
OLDI RESET

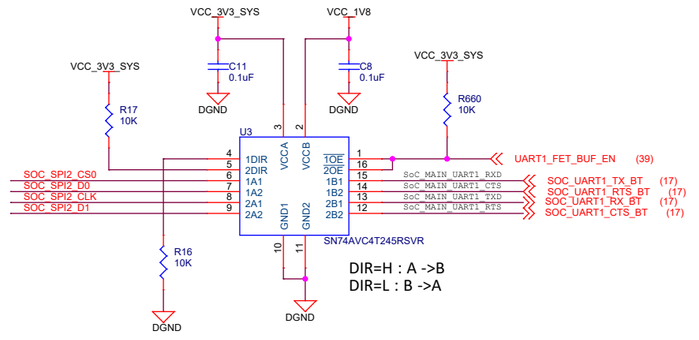




SoC UART1 FET SWITCH & BUFFER







OEn	SEL	INPUT/OUTPUT An	
L	H(DEFAULT)	An=nB2	FT4232
L	L	An=nB1	EXP CONN

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Title	OLDI DISPLAY INTERFACE	
Size C	PROC114E3	Rev E3
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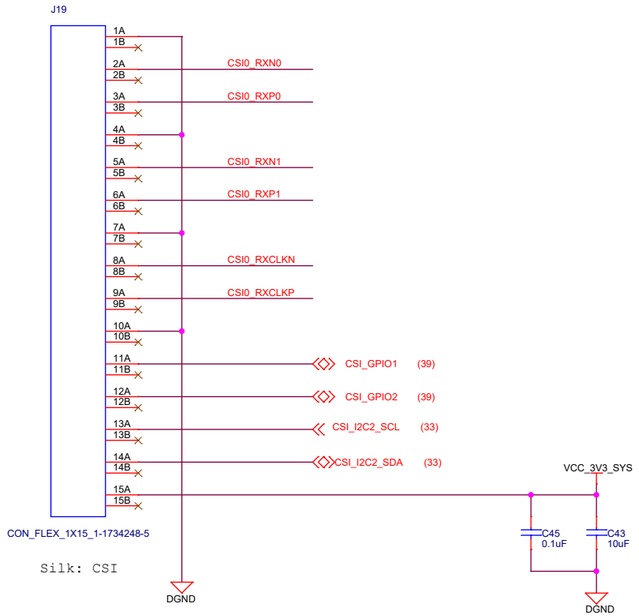
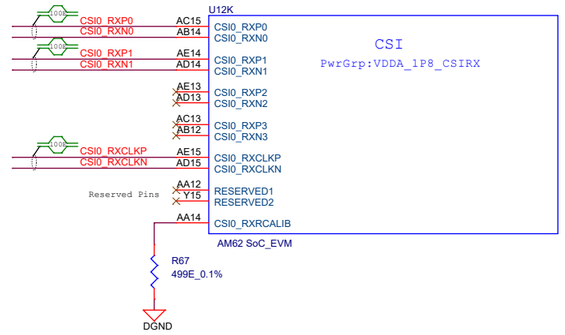
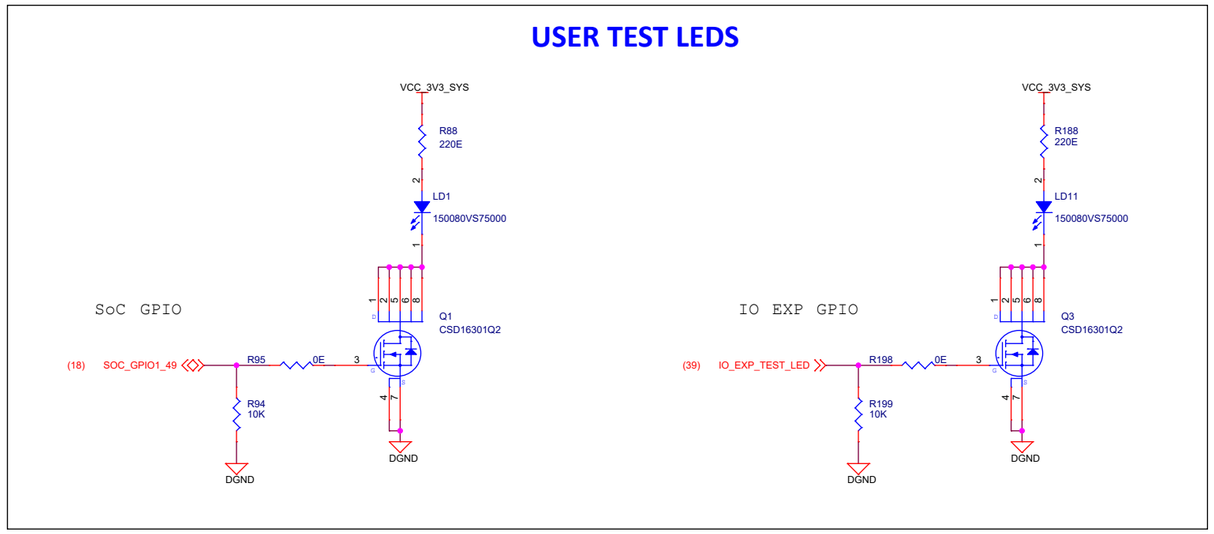
CSI INTERFACE

CSI CAMERA HEADER





USER TEST LEDS



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Title	CSI INTERFACE & USER TEST LEDS		
Size	PROC114E3		Rev
C			E3
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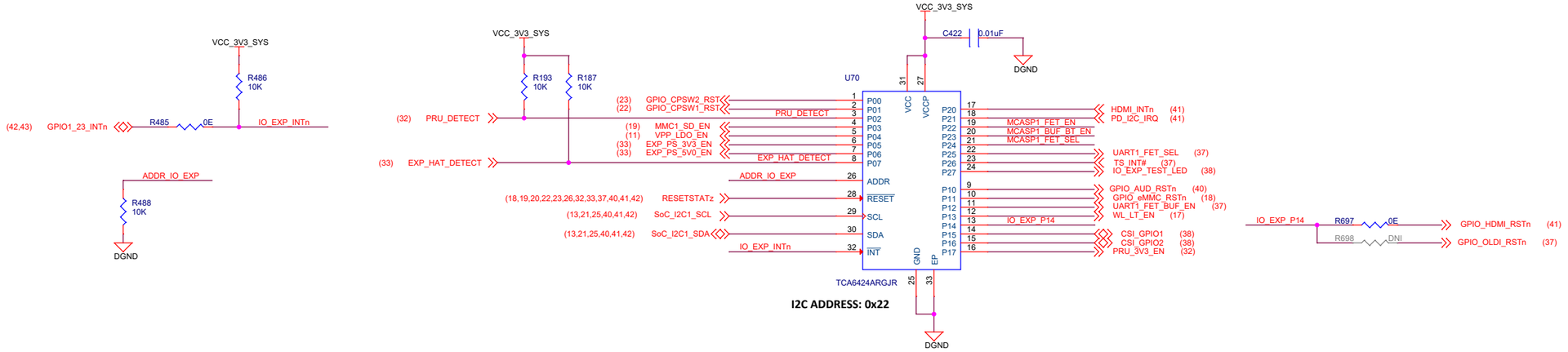
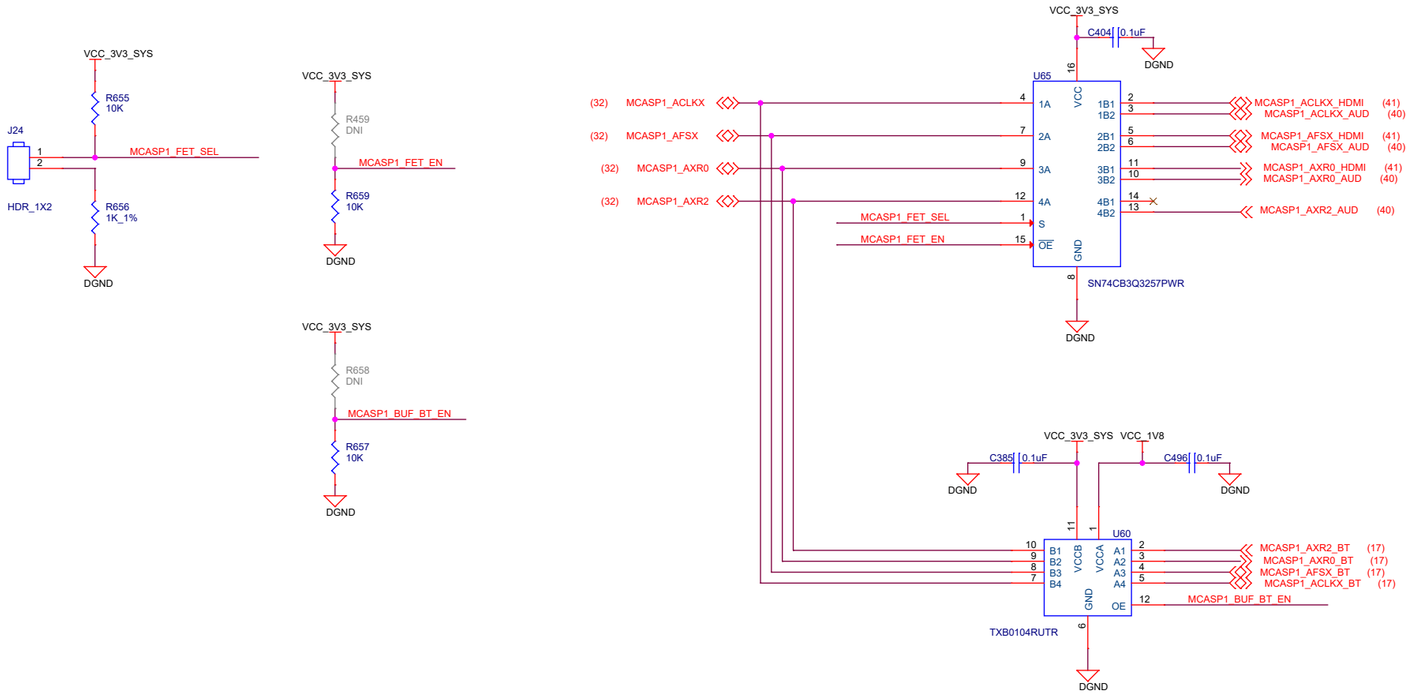
IO EXPANDER



MCASP1 FET SWITCH & BUFFER



OEn	SEL	INPUT/OUTPUT An	
L	H(DEFAULT)	An=nB2	MCASP1 - CODEC
L	L	An=nB1	MCASP1 - HDMI

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Title	IO EXPANDER		
Size: C	PROC114E3		Rev: E3
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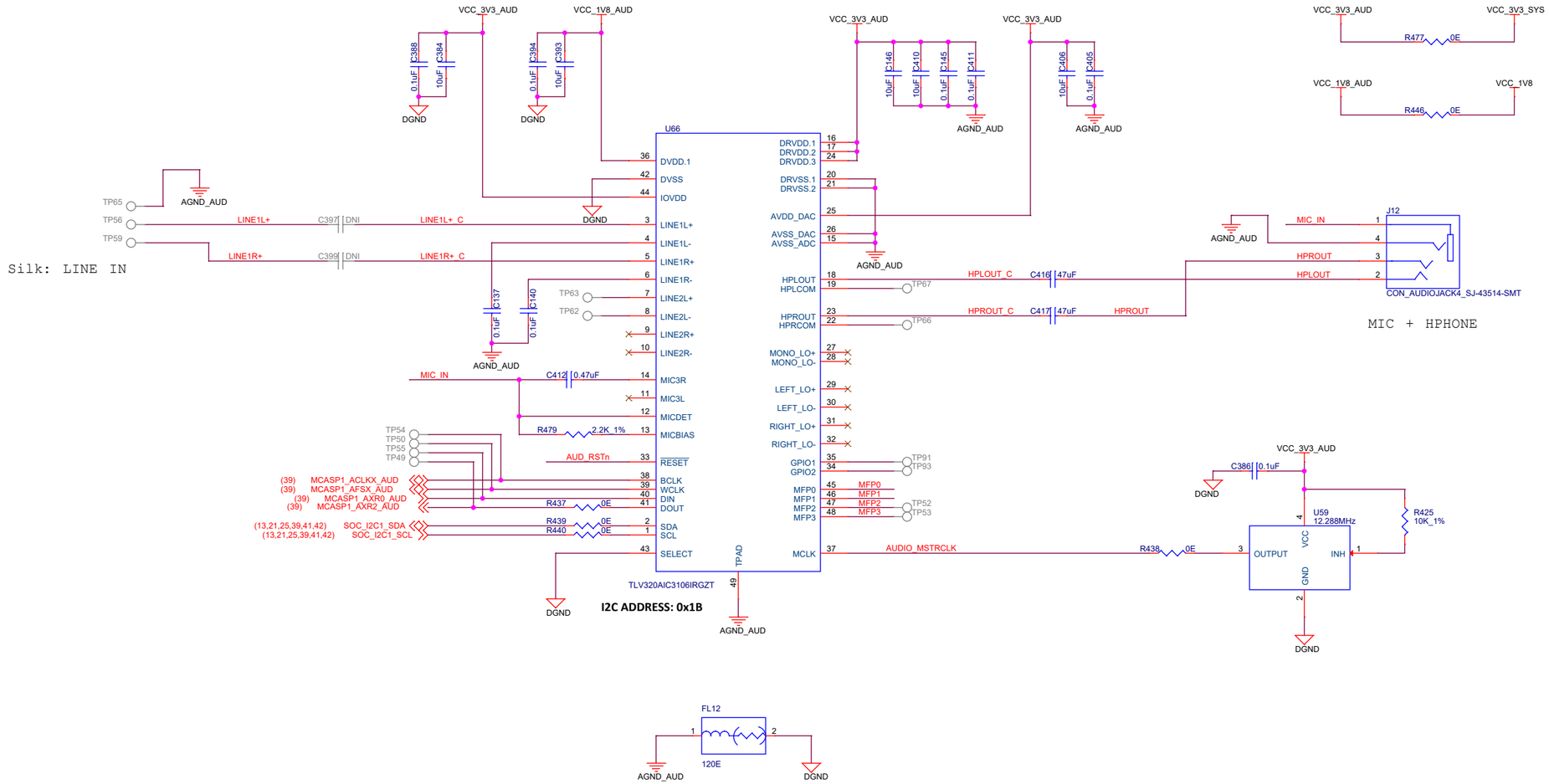
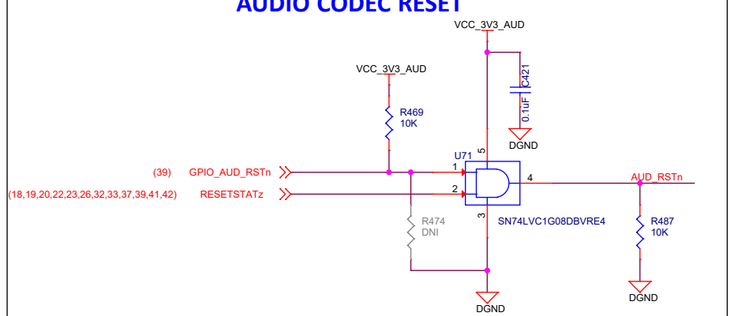
AUDIO CODEC

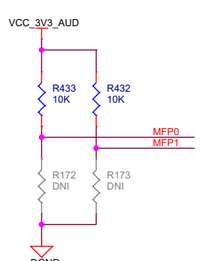


AUDIO CODEC RESET



CODEC I2C ADDRESS SELECTION

MFP0	MFP1	Device Address
0	0	0x18
0	1	0x19
1	0	0x1A
1	1	0x1B



Designed for TI by Mistral Solutions Pvt Ltd

Title	AUDIO CODEC	
Size C	PROC114E3	Rev E3
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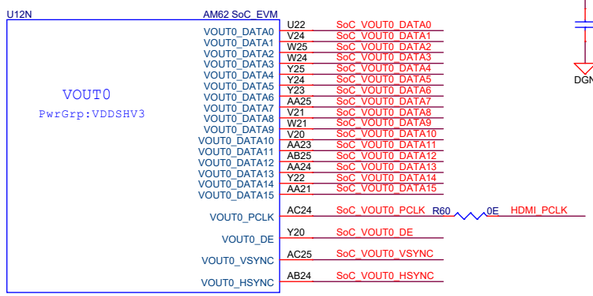
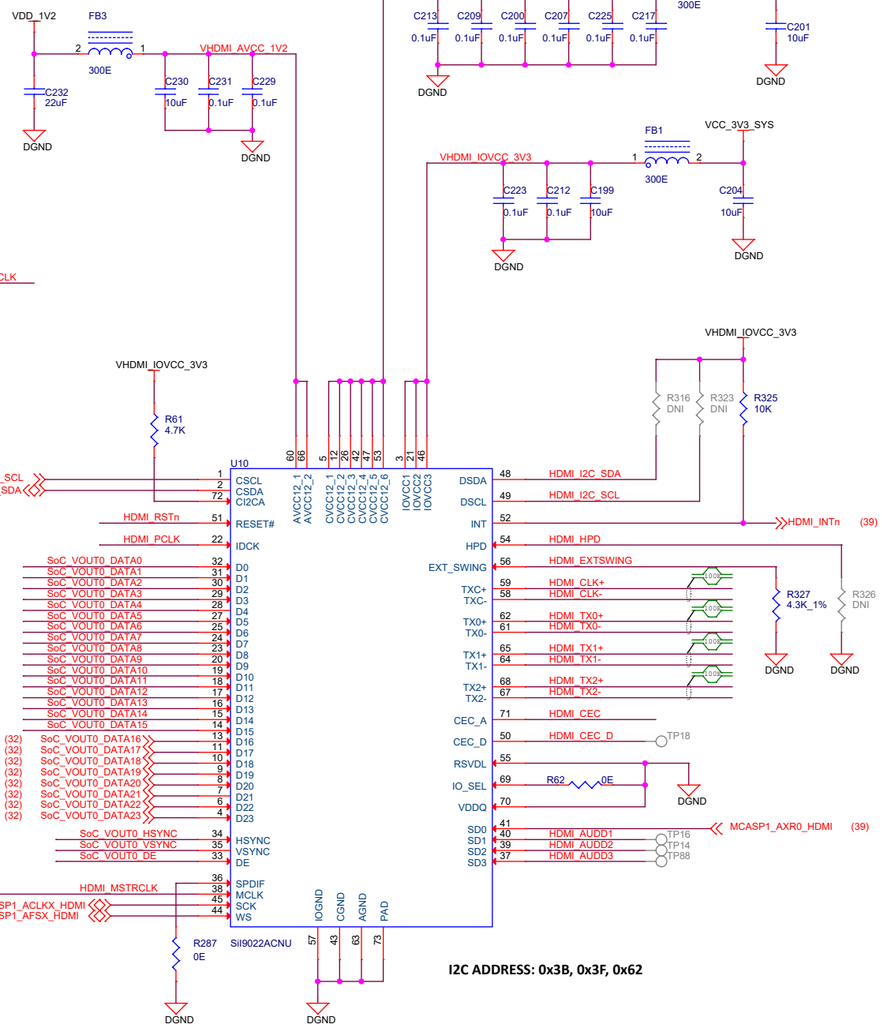
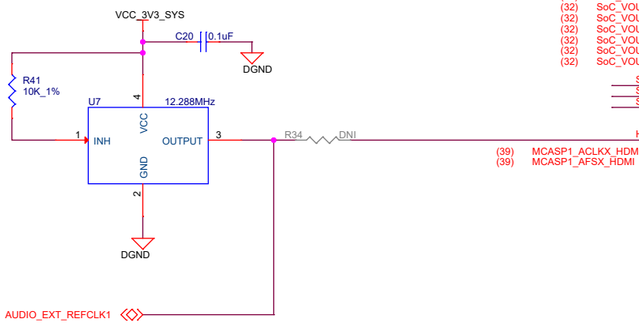
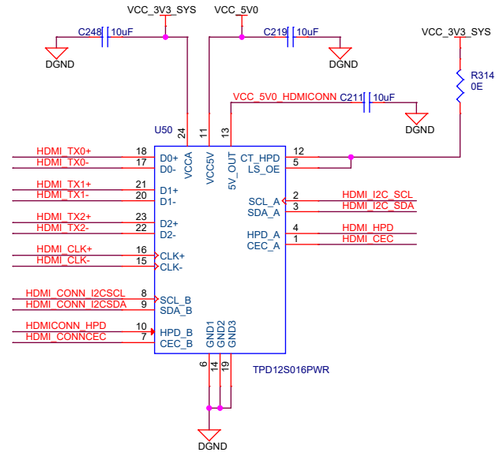
HDMI INTERFACE





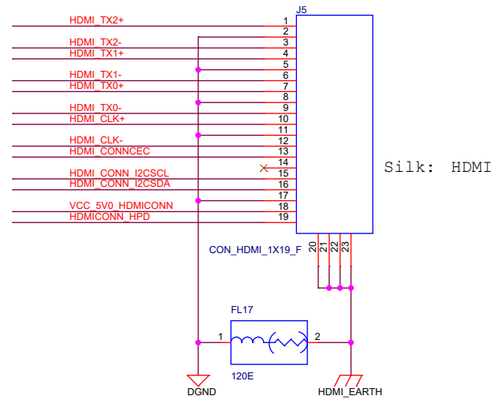


HDMI ESD DEVICE

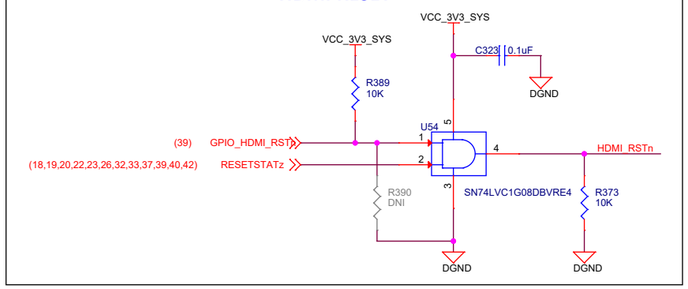


NOTE:
TPD12S016PWR has integrated pullup or pulldown resistors on the I2C and HPD lines hence no external pullup or pulldown required.

HDMI CONNECTOR



HDMI RESET



I2C ADDRESS: 0x3B, 0x3F, 0x62

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Title	HDMI INTERFACE		
Size	PROC114E3		Rev
C			E3
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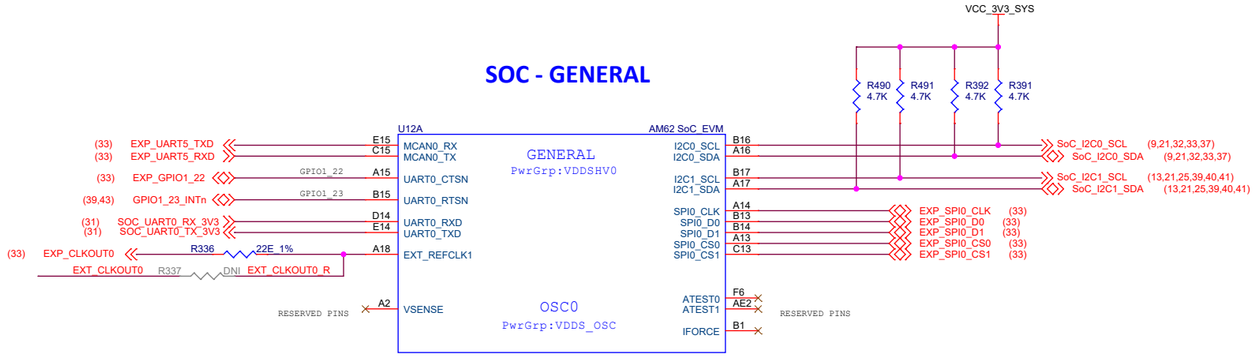
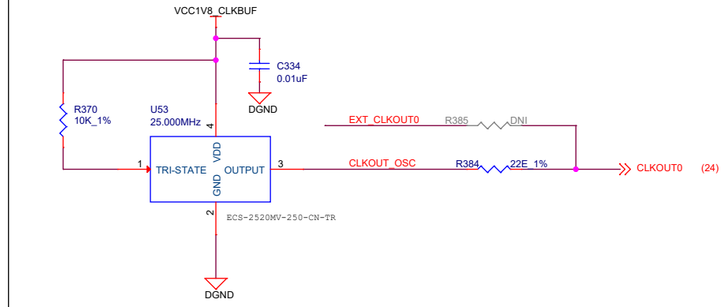
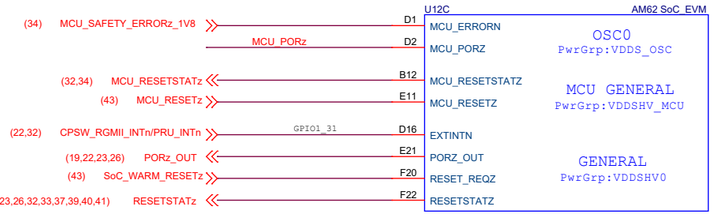
SOC - GENERAL

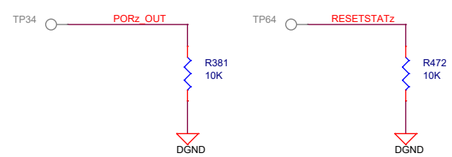


OSCILLATOR

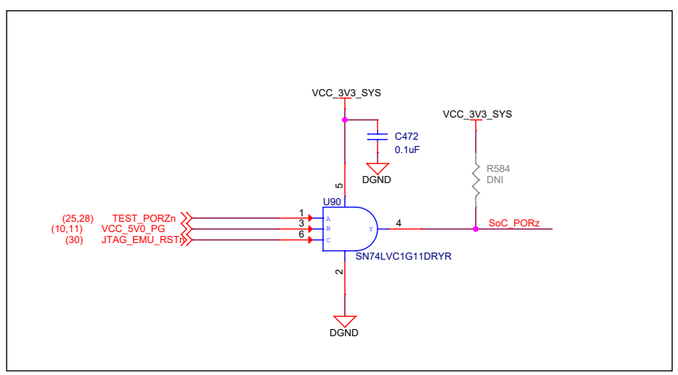


SOC - RESET

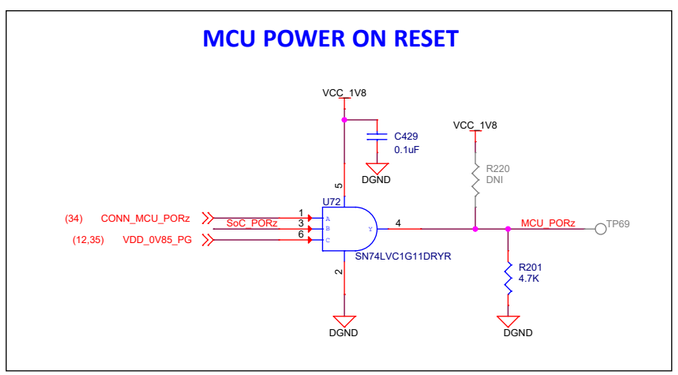




Pull-down resistor on PORz_OUT is provided to keep the signal low until the processor is released from reset during the power-up seqence



MCU POWER ON RESET



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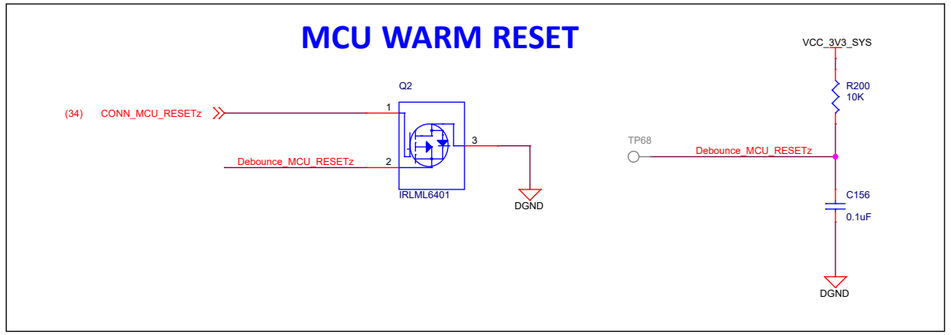
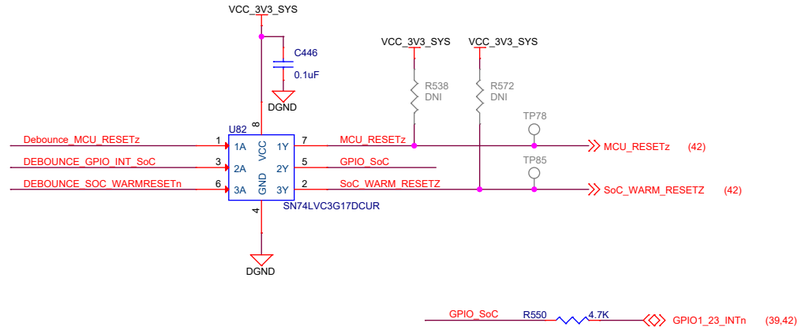
RESET

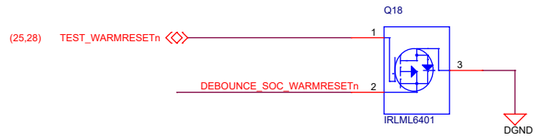
MCU WARM RESET

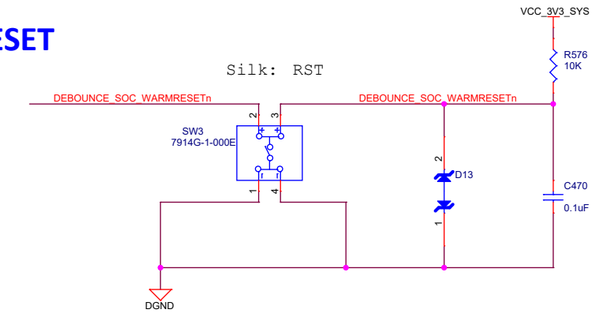


DEBOUNCE CIRCUIT

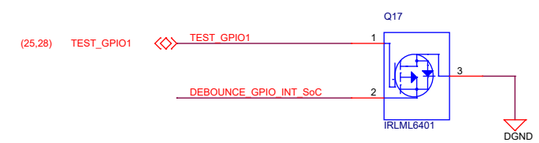


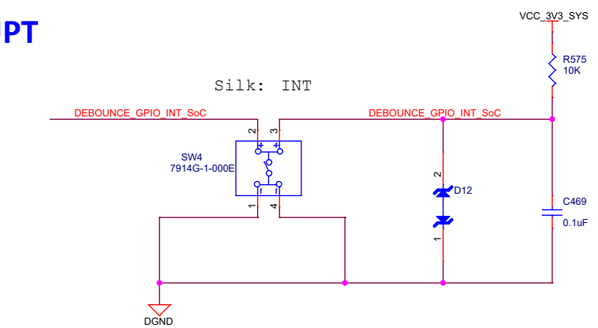
SOC WARM RESET





USER INTERRUPT





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HARDWARE SCHEMATICS

ASSEMBLY NOTES

1. All MSL components should be baked as per JEDEC standard.

2. PCB should be baked at 120 degree for 8 hours.

3. Board assembly must comply with workmanship standards. IPC-A-610 Class 2, unless otherwise specified.

4. These assemblies are ESD sensitive, ESD precautions shall be observed.

5. These assemblies must be clean and free from flux and all contaminants. Use of no clean flux is not acceptable.

6. Provide serial numbers to the assembled boards for identification.

7. The assembled board are wrapped in ESD Covers(individual) and packed securely before shipment.

BARE PCB

PROC114

PROC114E1

LOGOs

PCB LOGO DNI
Texas Intruments

PCB LOGO DNI
For Evaluation only; not FCC approved for resale

PCB LOGO DNI
WEEE Mark

PCB LOGO DNI
CE Mark

LABELS

Board Serial No.

LBL1
PCB LABEL
DNI

Assembly Revision

LBL2
PCB LABEL
DNI

FIDUCIALS

FID1 DNI
FID2 DNI
FID3 DNI
FID4 DNI
FID5 DNI
FID6 DNI

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